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Sasaki et al.

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(54) **ENERGY STORAGE APPARATUS**

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(58) **Field of Classification Search**
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See application file for complete search history.

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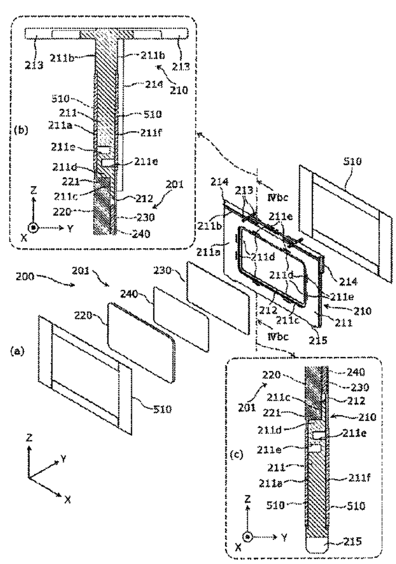
(57) **ABSTRACT**

An energy storage apparatus includes an energy storage device and a spacer disposed adjacent to the energy storage device in a first direction, in which the energy storage device has a case first surface on which an electrode terminal is disposed, a case second surface adjacent to the case first surface in a second direction intersecting the first direction, and a convex portion (gasket) protruding from the case first surface, and the spacer has a first protrusion that faces the convex portion in the second direction and protrudes along the convex portion without having a portion that faces the case second surface.

19 Claims, 10 Drawing Sheets



(51) **Int. Cl.**

H01M 50/55	(2021.01)
H01M 50/553	(2021.01)
H01M 50/562	(2021.01)

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Fig. 1

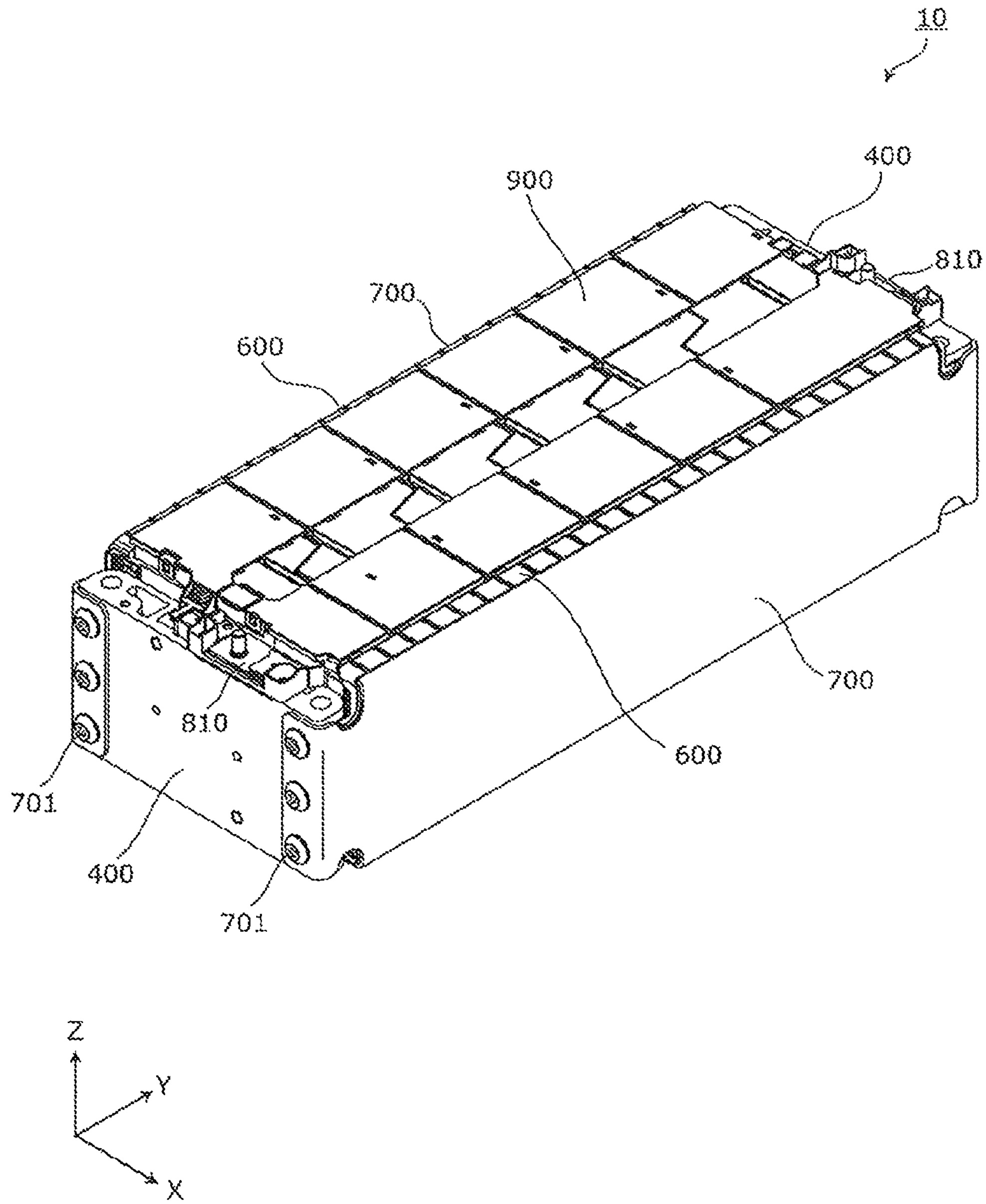

Fig. 2

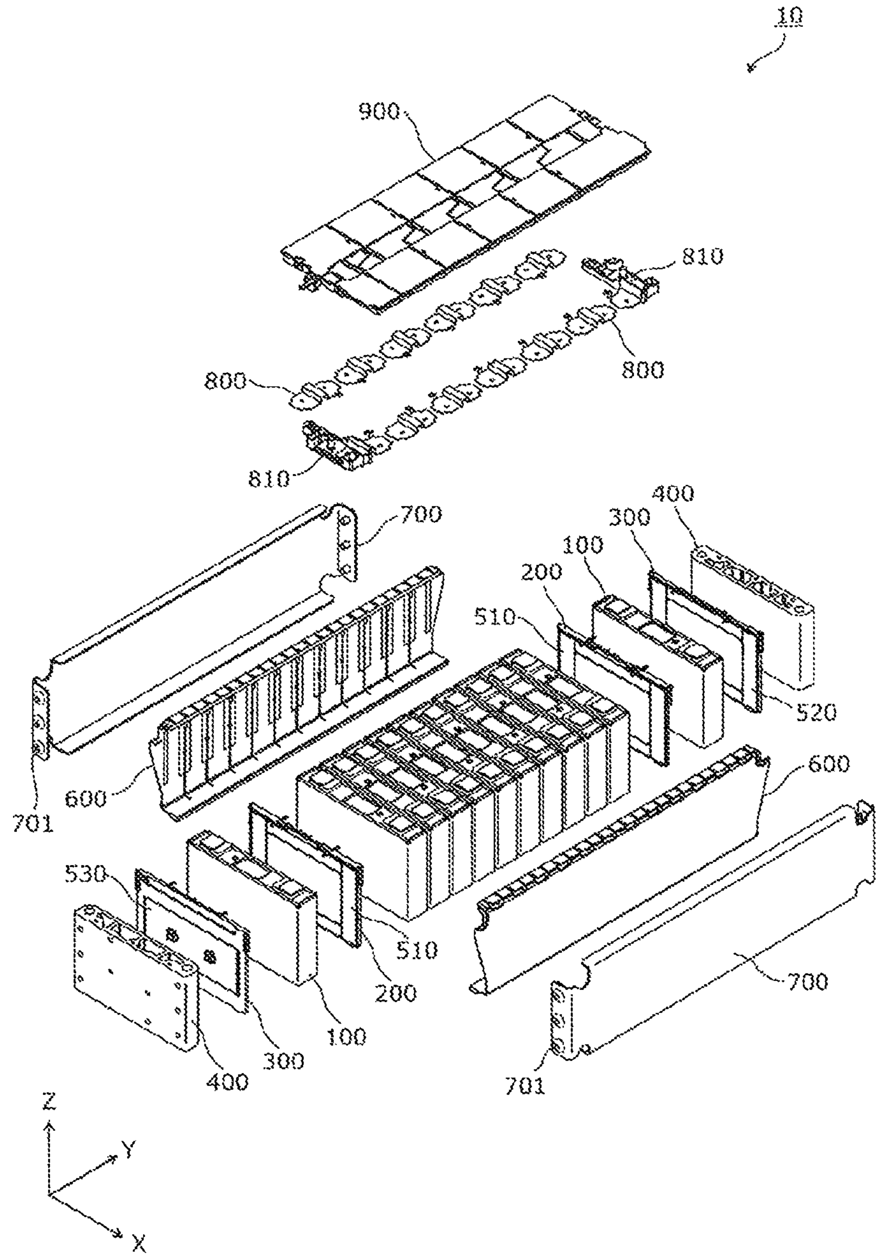
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Fig. 3

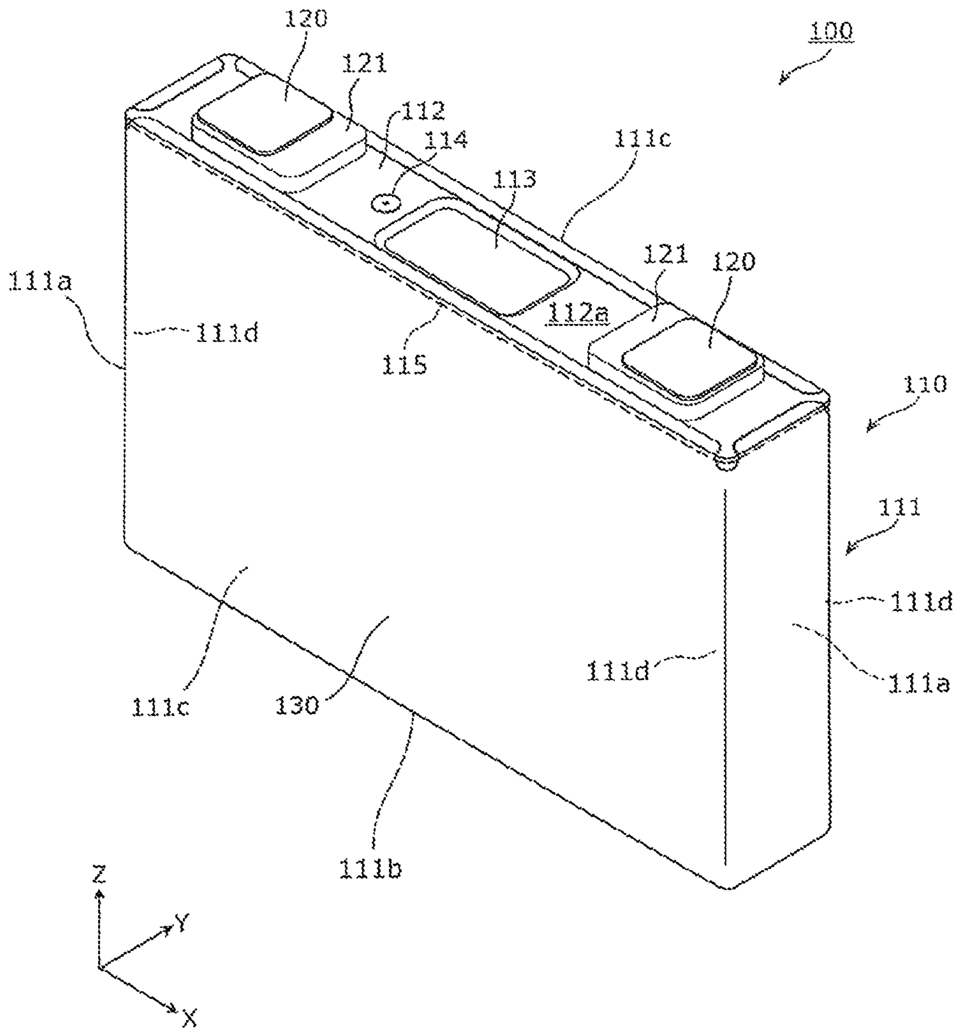

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Fig. 4

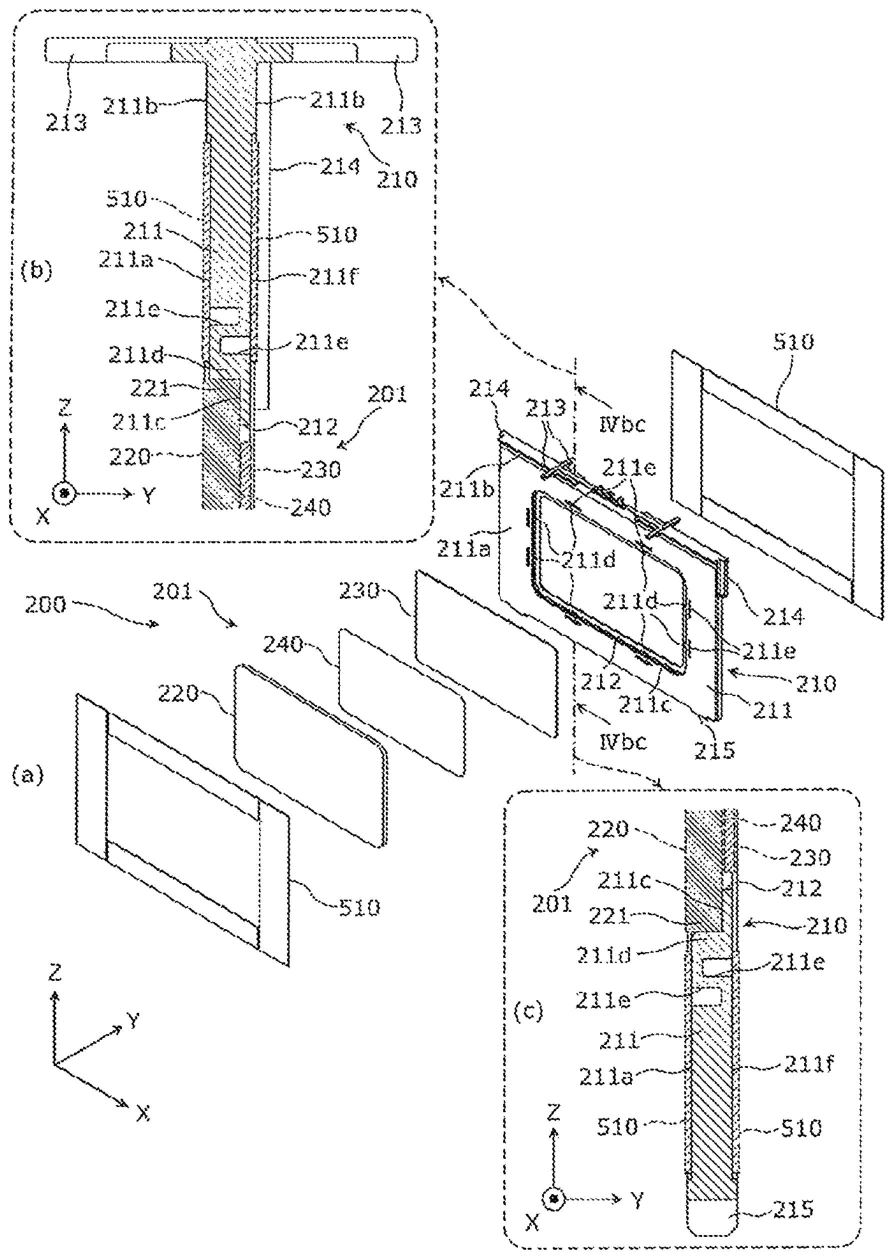

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Fig. 5

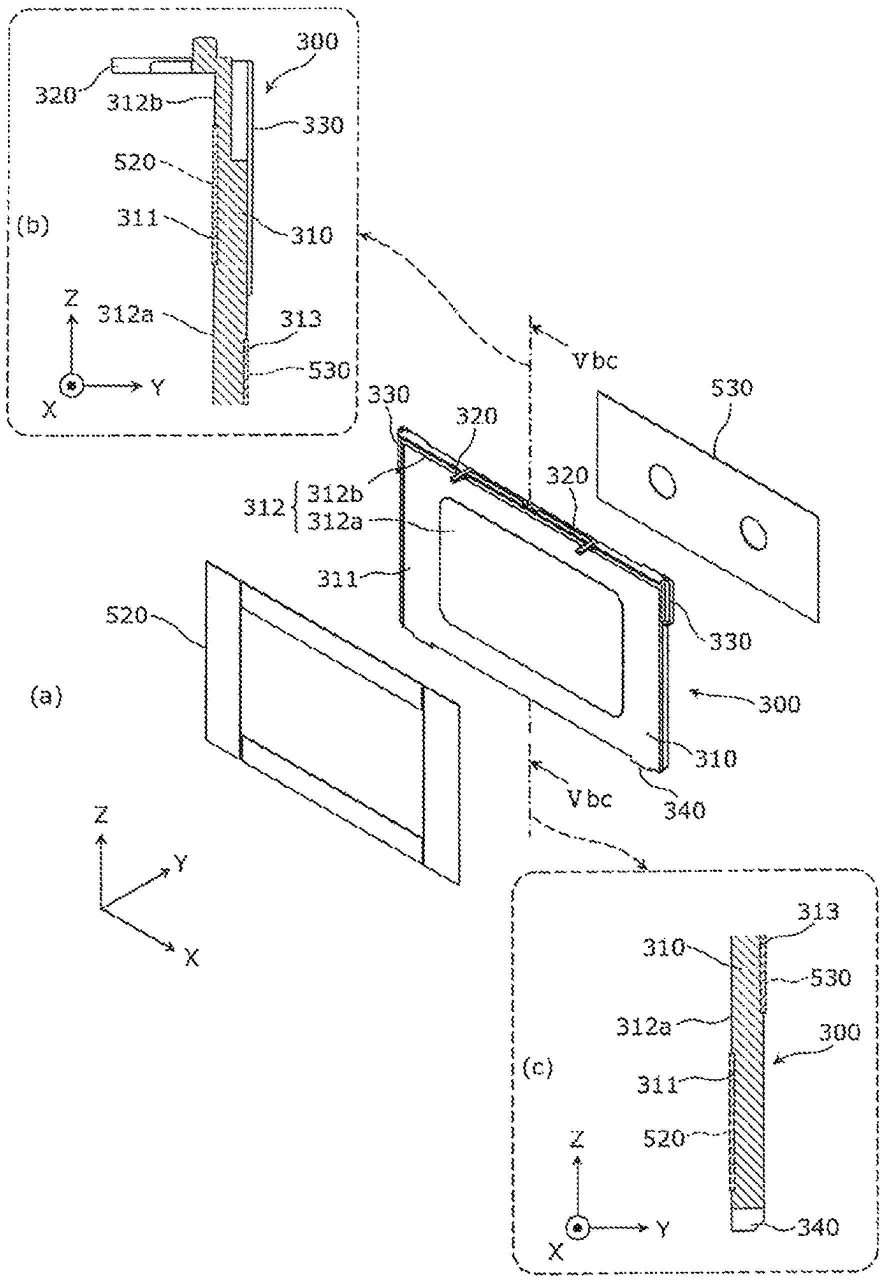

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Fig. 6

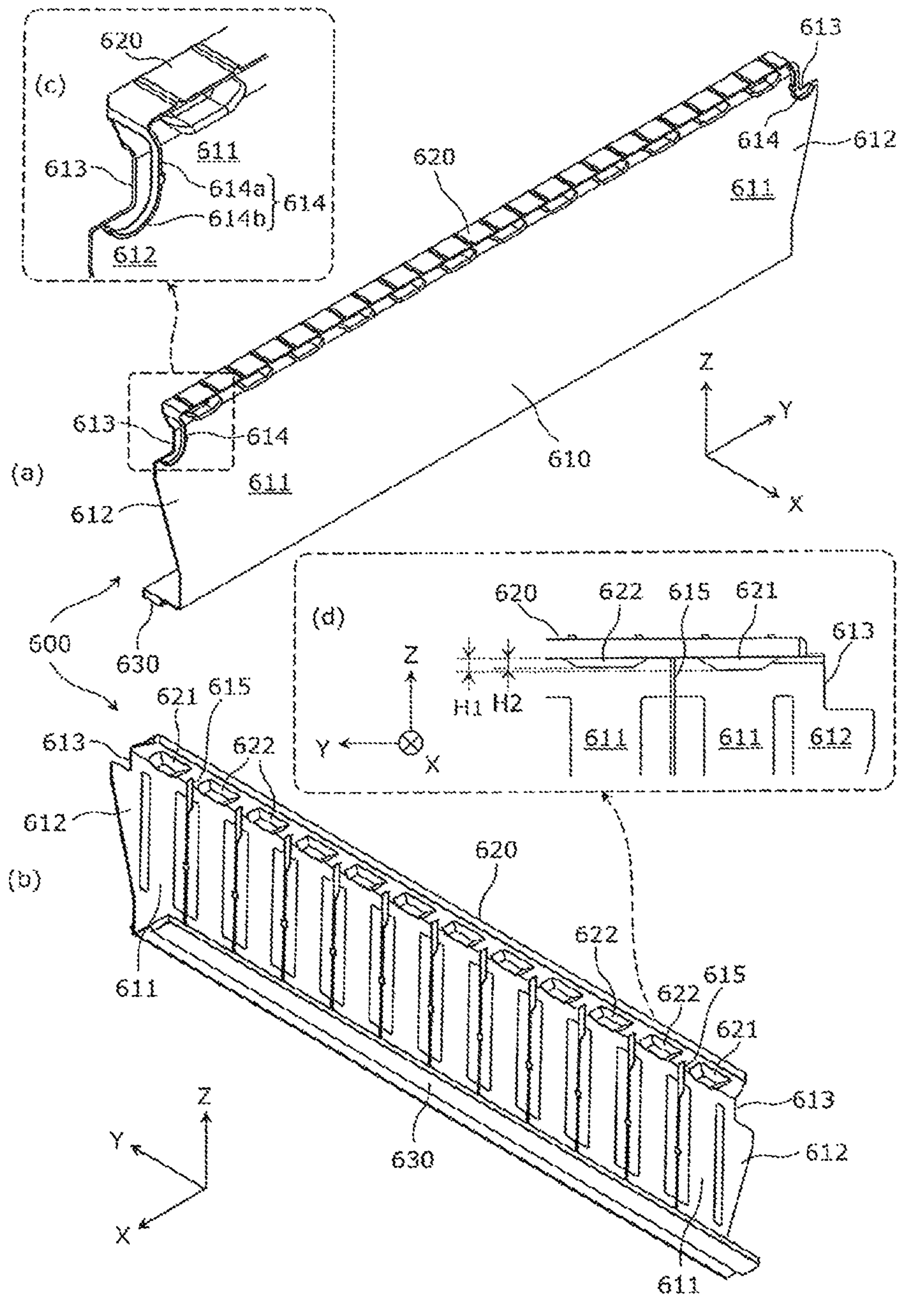

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Fig. 7

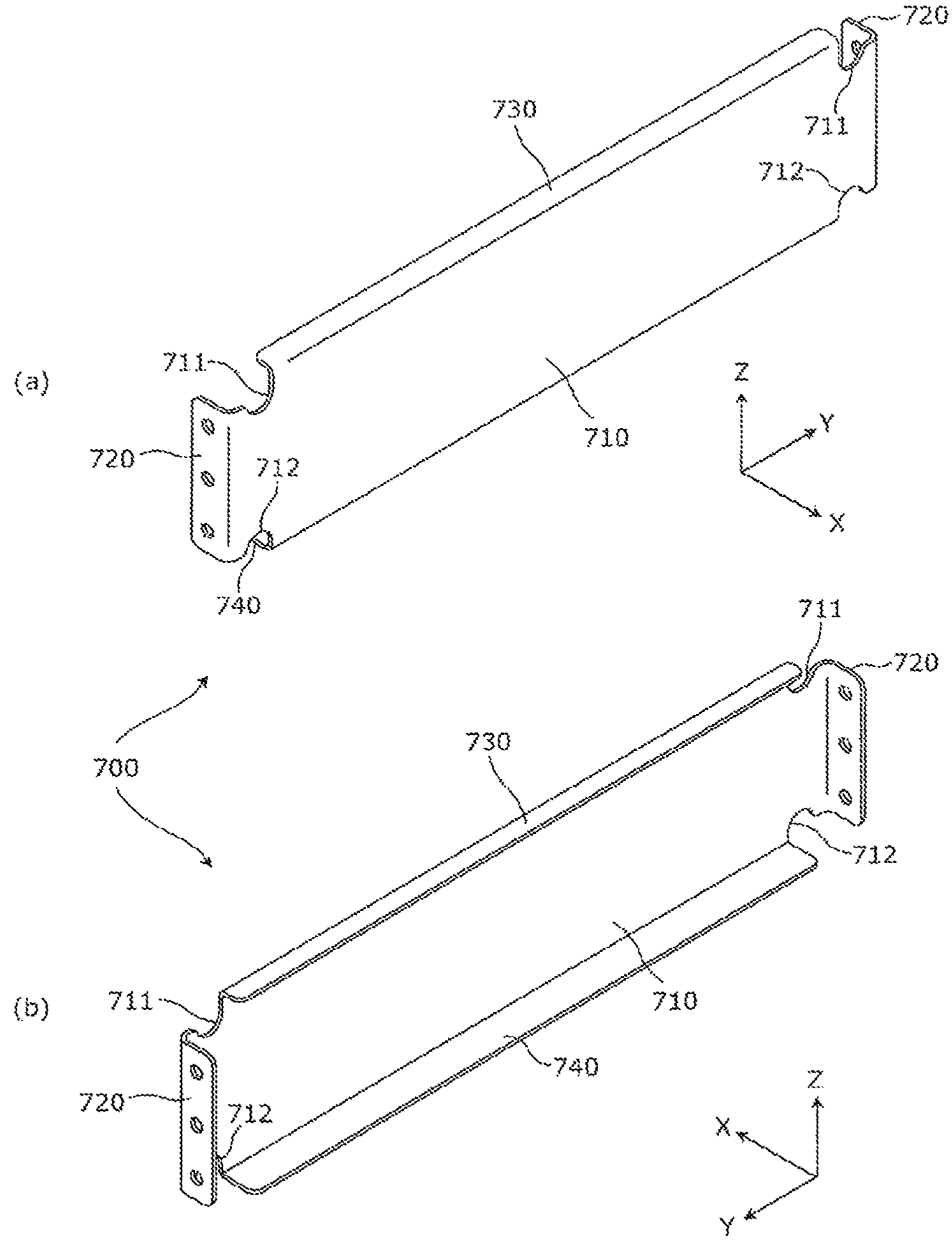

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Fig. 8

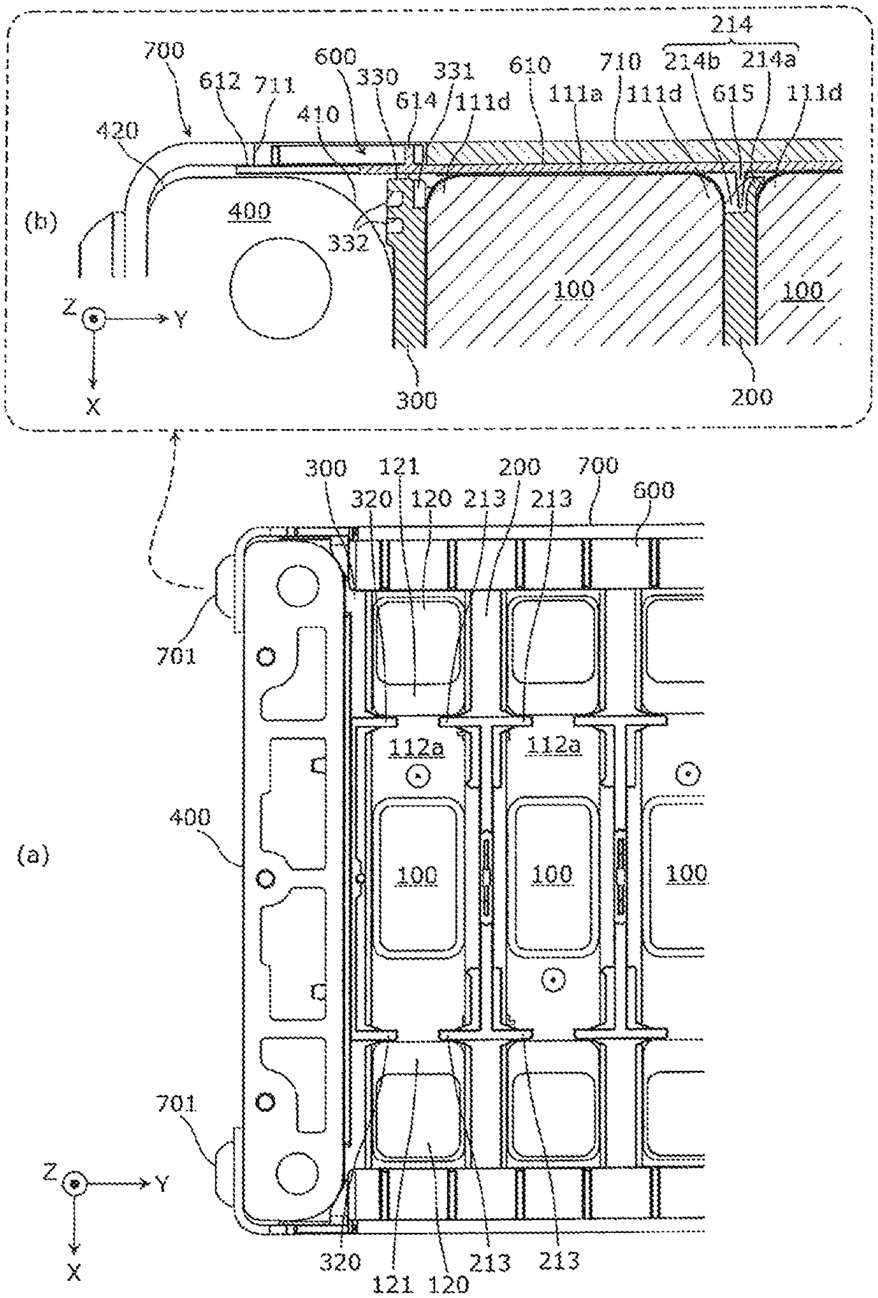

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Fig. 9

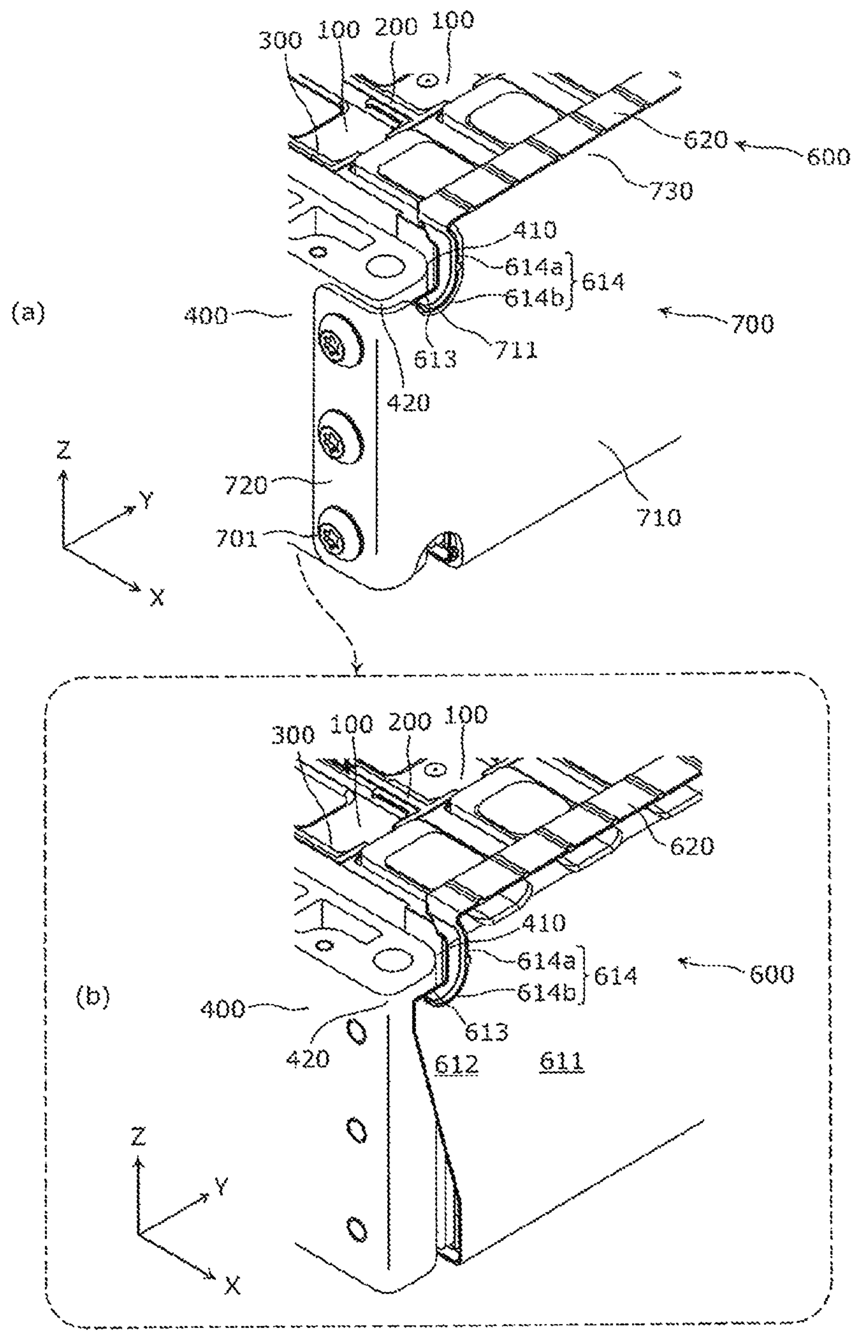
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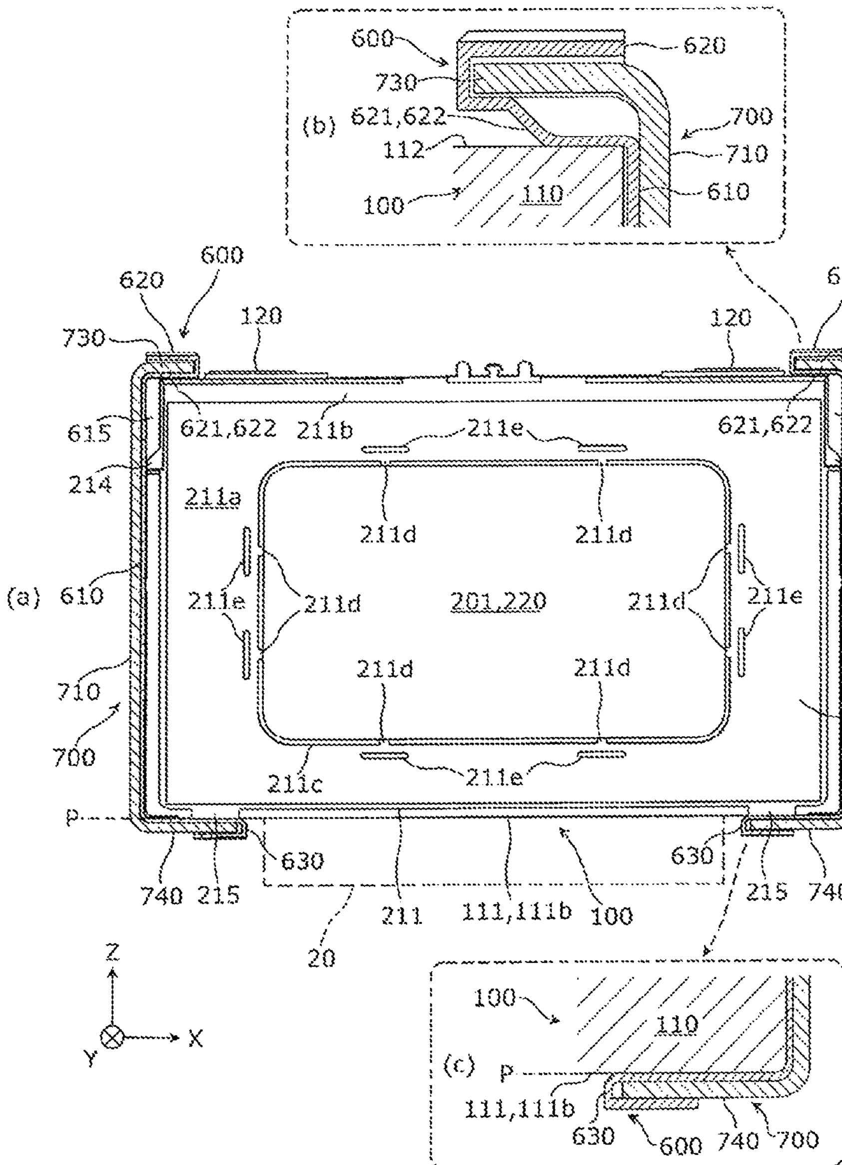
Fig. 10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ENERGY STORAGE APPARATUS

TECHNICAL FIELD

The present invention relates to an energy storage apparatus which includes an energy storage device and a spacer.

BACKGROUND ART

Conventionally, an energy storage apparatus including an energy storage device and a spacer is widely known. Patent Document 1 discloses an energy storage apparatus (battery module) including an energy storage device (battery cell) and a spacer, in which the spacer is configured to have side wall portions facing both side surfaces of the energy storage device.

PRIOR ART DOCUMENT

Patent Document

Patent Document 1: JP-A-2015-5362

SUMMARY OF THE INVENTION

Problems to be Solved by the Invention

The above-described energy storage apparatus with the conventional configuration cannot achieve downsizing.

An object of the present invention is to provide an energy storage apparatus capable of achieving downsizing.

Means for Solving the Problems

An energy storage apparatus according to one aspect of the present invention is an energy storage apparatus that includes an energy storage device and a first spacer disposed adjacent to the energy storage device in a first direction, and the energy storage device has a first surface on which an electrode terminal is disposed, a second surface adjacent to the first surface in a second direction intersecting the first direction, and a convex portion protruding from the first surface, and the first spacer has a protrusion that faces the convex portion in the second direction and protrudes along the convex portion without having a portion that faces the second surface.

The present invention can be realized not only as an energy storage apparatus described above but also as a spacer included in the energy storage apparatus.

Advantages of the Invention

According to the energy storage apparatus of the present invention, downsizing can be achieved.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. **1** is a perspective view showing an external appearance of an energy storage apparatus according to an embodiment.

FIG. **2** is an exploded perspective view showing components when the energy storage apparatus according to the embodiment is disassembled.

FIG. **3** is a perspective view showing a configuration of an energy storage device according to the embodiment.

FIG. **4** is a perspective view and a cross-sectional view showing a configuration of a spacer (intermediate spacer) according to the embodiment.

FIG. **5** is a perspective view and a cross-sectional view showing a configuration of a spacer (end spacer) according to the embodiment.

FIG. **6** is a perspective view and a plan view showing a configuration of an insulator according to the embodiment.

FIG. **7** is a perspective view showing a configuration of a side plate according to the embodiment.

FIG. **8** is a plan view and a cross-sectional view showing a positional relationship among the energy storage device, the spacer, an end member, the insulator, and the side plate according to the embodiment.

FIG. **9** is a perspective view showing a positional relationship among the energy storage device, the spacer, the end member, the insulator, and the side plate according to the embodiment.

FIG. **10** is a cross-sectional view showing a positional relationship among the energy storage device, the spacer, the insulator, and the side plate according to the embodiment.

MODE FOR CARRYING OUT THE INVENTION

The above-described energy storage apparatus with the conventional configuration cannot achieve downsizing. That is, in Patent Document 1 described above, the spacer aims at positioning with the energy storage device by sandwiching both side surfaces of the energy storage device with the side wall portions, but since the spacer has the side wall portion, and thus the width of the energy storage apparatus becomes large, downsizing of the energy storage apparatus cannot be achieved.

An energy storage apparatus according to one aspect of the present invention is an energy storage apparatus that includes an energy storage device and a first spacer disposed adjacent to the energy storage device in a first direction, and the energy storage device has a first surface on which an electrode terminal is disposed, a second surface adjacent to the first surface in a second direction intersecting the first direction, and a convex portion protruding from the first surface, and the first spacer has a protrusion that faces the convex portion in the second direction and protrudes along the convex portion without having a portion that faces the second surface.

According to this, in the energy storage apparatus, the energy storage device includes the first surface on which the electrode terminal is disposed, the second surface that is adjacent to the first surface and on the second direction side, and the convex portion protruding from the first surface. The first spacer has the protrusion that faces the second direction side of the convex portion and protrudes along the convex portion without having the portion that faces the second surface. In this way, by forming the protrusion protruding along the convex portion of the energy storage device on the first spacer, the energy storage device and the first spacer can be positioned without provision of a portion for sandwiching the side surface of the energy storage device on the first spacer. Accordingly, the width of the energy storage apparatus can be prevented from increasing, and downsizing of the energy storage apparatus can be achieved.

A width of the first spacer in the second direction may be smaller than a width of the energy storage device.

According to this, the width of the first spacer in the second direction is formed to be smaller than that of the energy storage device. By thus forming the width of the first spacer to be smaller than that of the energy storage device, it is possible to prevent the width of the energy storage apparatus from increasing and to achieve downsizing of the energy storage apparatus.

The protrusion may extend in a long shape along the convex portion.

According to this, the protrusion of the spacer extends in a long shape along the convex portion of the energy storage device. That is, since the protrusion of the spacer is not for insulating the first surface of the energy storage device but for restricting the movement of the convex portion of the energy storage device, it is possible to form the protrusion in a long shape without the need for increasing the width in the second direction. In this way, since the long-shaped protrusion may be formed on the first spacer, it is possible to achieve downsizing of the energy storage apparatus while easily producing the first spacer and reducing the amount of material used for the first spacer.

A protrusion amount of the protrusion may be smaller than half a width of the first surface in the first direction.

According to this, the protrusion amount of the protrusion of the first spacer is formed to be smaller than half the width of the first surface of the energy storage device in the first direction. In this way, by reducing the protrusion amount of the protrusion of the first spacer, it is possible to prevent the protrusion from interfering with other portions. By reducing the protrusion amount of the protrusion of the first spacer, it is possible to easily produce the first spacer, reduce the amount of material used for the first spacer, and prevent the protrusion from breaking.

Furthermore, a second spacer that sandwiches the energy storage device with the first spacer may be provided, and the second spacer may have another protrusion which faces the convex portion in the second direction and protrudes along the convex portion without having a portion that faces the second surface.

According to this, the energy storage apparatus includes the second spacer that sandwiches the energy storage device with the first spacer, and the second spacer has another protrusion which faces the second direction side of the convex portion of the energy storage device and protrudes along the convex portion without having the portion that faces the second surface of the energy storage device. In this way, by forming another protrusion protruding along the convex portion of the energy storage device also on the second spacer, it is possible to position the energy storage device and the second spacer without provision of a portion for sandwiching the side surface of the energy storage device on the second spacer. Accordingly, the width of the energy storage apparatus can be prevented from increasing, and downsizing of the energy storage apparatus can be achieved.

The energy storage device may have the two convex portions, and the first spacer may have the two protrusions arranged along the two convex portions, between the two convex portions or at positions sandwiching the two convex portions.

According to this, the energy storage device has the two convex portions, and the first spacer has the two protrusions arranged along the two convex portions between the two convex portions or at positions sandwiching the two convex portions. Thus, in the spacer, by arranging the two protrusions at positions sandwiched between the two convex portions of the energy storage device or at positions sandwiching the two convex portions, it is possible to position the energy storage device and the spacer in both directions in the arrangement direction of the two protrusions.

The convex portion is the electrode terminal disposed on the first surface, or a gasket disposed between the electrode terminal and a case of the energy storage device, and the protrusion may protrude along the electrode terminal or the gasket.

According to this, the convex portion of the energy storage device is the electrode terminal or the gasket, and the protrusion of the first spacer is disposed so as to protrude along the electrode terminal or the gasket. That is, the protrusion of the first spacer is disposed along the electrode terminal or the gasket of the energy storage device to position the energy storage device and the first spacer. As described above, by using the electrode terminal or the gasket of the energy storage device for positioning the energy storage device and the first spacer, it is not necessary to newly form a convex portion, so that the configuration is simplified and downsizing of the energy storage apparatus can be achieved.

Hereinafter, an energy storage apparatus according to an embodiment of the present invention is described with reference to the drawings. The embodiments described below show comprehensive or specific examples. However, numerical values, shapes, materials, components, arrangement positions and connection modes of the components, manufacturing processes, order of manufacturing processes, and the like described in the embodiments hereinafter are only examples and are not intended to limit the present invention. Among the components in the embodiments described hereinafter, the components which are not described in independent claims which describe uppermost concepts are described as arbitrary components. In each drawing, dimensions and the like are not strictly shown.

In the following description and drawings, the arrangement direction of a pair of electrode terminals in one energy storage device, the facing direction of a pair of short side surfaces in a case of one energy storage device, the arrangement direction of an insulator, the arrangement direction of a side plate, or the arrangement direction of an insulator and a side plate is defined as the X-axis direction. The arrangement direction of energy storage devices, the arrangement direction of spacers (intermediate spacers, end spacers), the arrangement direction of end members, the arrangement direction of energy storage devices, spacers, and end members, the facing direction of a pair of long side surfaces in a case of one energy storage device, or the thickness direction of the energy storage device, the spacer, or the end member is defined as the Y-axis direction. The arrangement direction of a case body and a lid of the energy storage device, the arrangement direction of the energy storage device, a bus bar, and a bus bar holding member, or the vertical direction is defined as the Z-axis direction. These X-axis direction, Y-axis direction, and Z-axis direction are directions that intersect each other (orthogonal in the present embodiment). Although the Z-axis direction may not be in the vertical direction depending on the usage mode, the Z-axis direction will be described below as the vertical direction for convenience of explanation. In the following description, the X-axis plus direction indicates the arrow direction of the X-axis, and the X-axis minus direction indicates the direction opposite to the X-axis plus direction. The same applies to the Y-axis direction and the Z-axis direction.

The Y-axis direction is an example of the first direction, and the X-axis direction is an example of the second direction. That is, the first direction is the direction in which the spacer is disposed with respect to the energy storage device, and is the Y-axis plus direction or the Y-axis minus direction. The second direction is a direction that intersects the first direction.

EMBODIMENT

[1 General Description of Energy Storage Apparatus **10**]

First, a configuration of an energy storage apparatus **10** is described. FIG. **1** is a perspective view showing an external appearance of the energy storage apparatus **10** according to the present embodiment. FIG. **2** is an exploded perspective view showing components when the energy storage apparatus **10** according to the present embodiment is disassembled.

The energy storage apparatus **10** is an apparatus which is charged with electricity from the outside or can discharge electricity to the outside. The energy storage apparatus **10** is a battery module (assembled battery) used for power storage application, power source application, or the like. Specifically, the energy storage apparatus **10** is used as a battery or the like for driving or starting engine of a moving body, such as an automobile such as an electric vehicle (EV), a hybrid electric vehicle (HEV), or a plug-in hybrid electric vehicle (PHEV), a motorcycle, a watercraft, a snowmobile, an agricultural machine, or a construction machine.

As shown in FIGS. **1** and **2**, the energy storage apparatus **10** includes a plurality of (twelve in the present embodiment) energy storage devices **100**, a plurality of spacers **200** and **300** (in the present embodiment, eleven spacers **200** and one pair of spacers **300**), a pair of end members **400**, a pair of insulators **600**, a pair of side plates **700**, a bus bar **800**, and a bus bar holding member **900**. A joining member **510** is disposed between the energy storage device **100** and the spacer **200**, a joining member **520** is disposed between the energy storage device **100** and the spacer **300**, and a joining member **530** is disposed between the spacer **300** and the end member **400**. A pair of external terminals **810** (a positive electrode external terminal and a negative electrode external terminal) that are terminals of the energy storage apparatus **10** are connected to the bus bar **800**. The energy storage apparatus **10** also includes a wiring for voltage measurement of the energy storage device **100**, a wiring for temperature measurement, a thermistor, and the like, but these are not shown and detailed description thereof is also omitted. The energy storage apparatus **10** may also include a circuit board for monitoring the charge state or discharge state of the energy storage device **100** or an electric device such as a relay.

The energy storage device **100** is a secondary battery (battery cell) which is charged with electricity or can discharge electricity, and more specifically, is a non-aqueous electrolyte secondary battery such as a lithium ion secondary battery. The energy storage device **100** has a flat rectangular parallelepiped shape (square shape) and is arranged adjacent to the spacers **200** and **300**. That is, each of the plurality of energy storage devices **100** is alternately arranged with each of the plurality of spacers **200** and **300**, and is arranged in the Y-axis direction. In the present embodiment, the eleven spacers **200** are arranged between the adjacent energy storage devices **100** among the twelve energy storage devices **100**, respectively. The pair of spacers **300** are arranged at positions sandwiching the energy storage devices **100** at the end portions among the twelve energy storage devices **100**.

The number of energy storage devices **100** is not limited to twelve and may be a number other than twelve. The shape of energy storage device **100** is not limited to a rectangular parallelepiped shape, and may be a polygonal prism shape other than a rectangular parallelepiped shape, or may be a laminate-type energy storage device. The energy storage device **100** is not limited to a non-aqueous electrolyte secondary battery, and may be a secondary battery other than the non-aqueous electrolyte secondary battery or may be a capacitor. The energy storage device **100** may be a primary battery that can use the electricity which is stored without the user having to charge the battery, instead of the secondary battery. The energy storage device **100** may be a battery using a solid electrolyte. The detailed description of a configuration of the energy storage device **100** is given later.

The spacers **200** and **300** are rectangular and plate-shaped spacers arranged lateral to (in the Y-axis plus direction or Y-axis minus direction) the energy storage device **100** for insulating the energy storage device **100** from other members. Specifically, the spacer **200** is an intermediate spacer disposed between the two adjacent energy storage devices **100** for insulating between the two energy storage devices **100**. More specifically, the joining members **510** are arranged on both sides of the spacer **200** in the Y-axis direction, and the joining members **510** join the spacer **200** and the energy storage devices **100** on both sides in the Y-axis direction. In the present embodiment, the eleven spacers **200** are arranged corresponding to the twelve energy storage devices **100**. However, when the number of energy storage devices **100** is other than twelve, the number of spacers **200** is also changed according to the number of energy storage devices **100**.

The spacer **300** is an end spacer that is disposed between the energy storage device **100** at the end portion and the end member **400** for insulating between the energy storage device **100** at the end portion and the end member **400**. Specifically, the joining member **520** is disposed on the energy storage device **100** side of the spacer **300**, and the spacer **300** and the energy storage device **100** are joined by the joining member **520**. The joining member **530** is disposed on the end member **400** side of the spacer **300**, and the spacer **300** and the end member **400** are joined by the joining member **530**. The detailed description of a configuration of these spacers **200** and **300** will be given later.

The end member **400** and the side plate **700** are members that press the energy storage device **100** from the outside in an arrangement direction (Y-axis direction) of the plurality of energy storage devices **100**. That is, the end member **400** and the side plate **700** sandwich the plurality of energy storage devices **100** from both sides in the arrangement direction, thereby pressing the respective energy storage devices **100** included in the plurality of energy storage devices **100** from both sides in the arrangement direction.

Specifically, the end members **400** are flat block-shaped end plates (holding members) arranged on both sides of the plurality of energy storage devices **100** in the Y-axis direction for sandwiching and holding the plurality of energy storage devices **100** from both sides in the arrangement direction (Y-axis direction) of the plurality of energy storage devices **100**. The end member **400** is formed of a metal (conductive) member such as steel or stainless steel from the viewpoint of strength. The material of the end member **400** is not particularly limited, and may be formed of a high-strength insulating member or may be subjected to an insulating treatment. The end member **400** is an example of a lateral member disposed lateral to the energy storage device **100**.

The side plate **700** is a long and flat plate-shaped restraining member (restraint bar) having both ends attached to the end members **400** for restraining the plurality of energy storage devices **100**. That is, the side plate **700** is disposed extending in the Y-axis direction so as to straddle the plurality of energy storage devices **100** and the plurality of spacers **200** and **300**, for applying a restraining force in the arrangement direction of these (Y-axis direction) with respect to the plurality of energy storage devices **100** and the plurality of spacers **200** and **300**. In the present embodiment, on both sides of the plurality of energy storage devices **100** in the X-axis direction, two side plates **700** are arranged at positions sandwiching the insulators **600** with the energy storage devices **100** (specifically, case second surfaces **111***a* described later). Each of the two side plates **700** is attached to the end portions of the two end members **400** in the X-axis direction at both ends in the Y-axis direction. Accordingly, the two side plates **700** sandwich and restrain the plurality of energy storage devices **100** and the plurality of spacers **200** and **300** from both sides in the X-axis direction and both sides in the Y-axis direction.

The side plate **700** is fixed to the end member **400** by a plurality of fixing members **701** arranged in the Z-axis direction. In the present embodiment, the fixing member **701** is a bolt that penetrates the side plate **700** and is joined to the end member **400**. The attachment of the side plate **700** to the end member **400** is not limited to fixing with the bolt, and may be joint by welding, adhesion, or the like. Like the end member **400**, the side plate **700** is a conductive member formed of a metal (conductive) member such as steel or stainless steel from the viewpoint of strength, but may be formed of a high-strength insulating member, or may be subjected to an insulating treatment. The side plate **700** is an example of a conductive member that sandwiches an insulating member with the energy storage device **100**, or an outer portion that is disposed outside a pressing member that presses the energy storage device **100**. The detailed description of a configuration of the side plate **700** will be given later.

The insulator **600** is a long and flat plate-shaped insulating member that is disposed on both sides of the plurality of energy storage devices **100** in the X-axis direction and extended in the Y-axis direction. That is, the insulator **600** is disposed between the plurality of energy storage devices **100** and the plurality of spacers **200** and **300** and the side plate **700** so as to straddle the plurality of energy storage devices **100** and the plurality of spacers **200** and **300**, and insulates the energy storage devices **100** from the side plate **700**. The insulator **600** is formed of an insulating material, such as polycarbonate (PC), polypropylene (PP), polyethylene (PE), polyphenylene sulfide resin (PPS), polyethylene terephthalate (PET), polyether ether ketone (PEEK), tetrafluoroethylene/perfluoroalkyl vinyl ether (PFA), polytetrafluoroethylene (PTFE), polybutylene terephthalate (PBT), poly ether sulfone (PES), ABS resin, ceramics, and composite materials thereof. The insulator **600** may be formed of any material as long as it is an insulating member, and the two insulators **600** may be formed of members of different materials.

The insulator **600** also has a function of pressing the plurality of energy storage devices **100** in the Z-axis minus direction. That is, the plurality of energy storage devices **100** have a configuration in which they are placed on a cooling device **20** (see FIG. **10**) and cooled, and the insulator **600** presses the plurality of energy storage devices **100** toward the cooling device **20**. The cooling device **20** is a device that cools the energy storage apparatus **10** (the plurality of energy storage devices **100**) by water cooling, for example. The energy storage apparatus **10** has a configuration in which the energy storage apparatus **10** is mounted not on the cooling device **20** but on the vehicle body of the automobile on which the energy storage apparatus **10** is mounted, or on an outer case or the like that houses the energy storage apparatus **10**. The insulator **600** may be configured to press the plurality of energy storage devices **100** toward the vehicle body, the outer case, or the like. The insulator **600** is an example of an inside part disposed in the inside of a first insulating member disposed on the X-axis direction side of the energy storage device **100**, an abutting member that abuts on the energy storage device **100**, or the pressing member that presses the energy storage device **100**. The detailed description of a configuration of the insulator **600** will be given later.

The bus bar **800** is a conductive plate-shaped member that is disposed on the plurality of energy storage devices **100** for electrically connecting the electrode terminals of the plurality of energy storage devices **100**. In the present embodiment, the bus bar **800** connects the plurality of energy storage devices **100** in series by connecting positive electrode terminals and negative electrode terminals of the adjacent energy storage devices **100** in order. The external terminals **810** for the positive and negative electrodes are connected to the bus bar **800** disposed at the end portions. The bus bar **800** is formed of a conductive member made of metal such as copper, copper alloy, aluminum, aluminum alloy. The connection mode of the energy storage device **100** is not particularly limited, and any of the energy storage devices **100** may be connected in parallel.

The bus bar holding member **900** is a plate-shaped member (busbar plate, busbar frame) that holds the bus bar **800** and other wirings (not shown), and can insulate the bus bar **800** and the like from the other members and regulate the position of the bus bar **800** and the like. Specifically, the bus bar holding member **900** has a main body portion and a lid portion, and is configured capable of housing the bus bar **800** and the like by opening the lid portion and placing the bus bar **800** and the like on the main body portion, and then closing the lid portion. The bus bar holding member **900** is formed of an insulating material such as PC, PP, PE, PPS, PET, PEEK, PFA, PTFE, PBT, PES, ABS resin, ceramics, and composite materials thereof.

[2 Detailed Description of Energy Storage Device **100**]

Next, the configuration of the energy storage device **100** is described in detail. FIG. **3** is a perspective view showing the configuration of the energy storage device **100** according to the present embodiment.

As shown in FIG. **3**, the energy storage device **100** includes a case **110**, two electrode terminals **120** (a positive electrode terminal and a negative electrode terminal), two gaskets **121**, and an insulating sheet **130**. Inside the case **110**, an electrode assembly, a current collector (a positive electrode current collector and a negative electrode current collector), an electrolytic solution (non-aqueous electrolyte), and the like are housed, but they are not shown. The electrolytic solution is not particularly limited in type as long as it does not impair the performance of the energy storage device **100**, and various electrolytic solutions can be selected. The gasket is also disposed between the case **110** (lid body **112** described later) and the current collector, and spacers are arranged at the sides of the current collector, but these are not shown.

The case **110** is a rectangular parallelepiped (square) case having a case main body **111** having an opening formed therein and a lid body **112** closing the opening of the case main body **111**. The lid body **112** is a rectangular plate-shaped member that constitutes a lid portion of the case **110**, and is disposed on the Z-axis plus direction side of the case main body **111**. The lid body **112** has a case first surface **112***a* on which the electrode terminals **120** are arranged. The case first surface **112***a* is a rectangular flat surface (outer surface or upper surface) disposed on the Z-axis plus direction side of the lid body **112** and extending in the X-axis direction. The lid body **112** is also provided with a gas release valve

113 that, when a pressure inside the case **110** rises, releases the pressure, a liquid injection part **114** for injecting an electrolytic solution into the inside of the case **110**, and the like.

The case main body **111** is a bottomed member constituting a main body portion of the case **110** and having a rectangular tubular shape, has two case second surfaces **111***a* on both side surfaces in the X-axis direction, has a case third surface **111***b* on the Z-axis minus direction side, and has two case fourth surfaces **111***c* on both side surfaces in the Y-axis direction. The case second surface **111***a* is a rectangular flat surface that forms a short side surface of the case **110**. In other words, the case second surface **111***a* is a surface adjacent to the case first surface **112***a*, the case third surface **111***b*, and the case fourth surface **111***c* and having a smaller area than the case fourth surface **111***c*. The case third surface **111***b* is a rectangular flat surface that forms the bottom surface of the case **110**. In other words, the case third surface **111***b* is a surface facing the case first surface **112***a* and adjacent to the case second surface **111***a* and the case fourth surface **111***c*. The case fourth surface **111***c* is a rectangular flat surface that forms a long side surface of the case **110**. In other words, the case fourth surface **111***c* is a surface adjacent to the case first surface **112***a*, the case second surface **111***a*, and the case third surface **111***b*, and having a larger area than the case second surface **111***a*. The case main body **111** has a curved case corner portion **111***d* at a boundary portion between the case second surface **111***a* and the case fourth surface **111***c*.

With such a configuration, the case **110** is configured such that after the electrode assembly and the like are housed inside the case main body **111**, the case main body **111** and the lid body **112** are joined by welding or the like to form a joint **115**, and thereby the inside is sealed. That is, on the side surfaces of the case **110** (the surfaces on both sides in the X-axis direction and both sides in the Y-axis direction), the joints **115**, in which the case main body **111** and the lid body **112** are joined, are formed. The material of the case **110** (the case main body **111** and the lid body **112**) is not particularly limited, but is preferably a weldable (joinable) metal such as stainless steel, aluminum, aluminum alloy, iron, and plated steel plate.

The electrode terminals **120** are terminals (a positive electrode terminal and a negative electrode terminal) of the energy storage device **100** disposed on the case first surface **112***a* of the case **110**, and are electrically connected to the positive electrode plate and the negative electrode plate of the electrode assembly through the current collector, respectively. That is, the electrode terminal **120** is a metal member for leading the electricity stored in the electrode assembly to the external space of the energy storage device **100**, and for introducing the electricity into the internal space of the energy storage device **100** to store the electricity in the electrode assembly. The electrode terminal **120** is formed of aluminum, aluminum alloy, copper, copper alloy, or the like.

The gasket **121** is a member disposed around the electrode terminal **120** and between the electrode terminal **120** and the lid body **112** of the case **110** for ensuring insulation and airtightness between the electrode terminal **120** and the case **110**. The gasket **121** is formed of an insulating material such as PP, PE, PPS, PET, PEEK, PFA, PTFE, PBT, PES, and ABS resin.

The electrode terminal **120** and the gasket **121** are convex portions protruding from the case first surface **112***a* of the case **110**. That is, the electrode terminal **120** or the gasket **121** is an example of a convex portion of the energy storage device **100**, and the energy storage device **100** has the two convex portions on both sides of the case first surface **112***a* in the X-axis direction. In the present embodiment, the energy storage device **100** has the gasket **121** as the convex portion.

The electrode assembly is an energy storage element (power generating element) formed by stacking a positive electrode plate, a negative electrode plate, and a separator. The positive electrode plate included in the electrode assembly has a positive active material layer formed on a positive electrode substrate layer, which is a long strip-shaped current collector foil made of a metal such as aluminum or an aluminum alloy. The negative electrode plate has a negative active material layer formed on a negative electrode substrate layer which is a long strip-shaped current collector foil made of a metal such as copper or a copper alloy. As the positive active material used for the positive active material layer and the negative active material used for the negative active material layer, known materials can be appropriately used as long as they can store and release lithium ions. The current collector is a member (a positive electrode current collector and a negative electrode current collector) having electrical conductivity and rigidity, which is electrically connected to the electrode terminal **120** and the electrode assembly. The positive electrode current collector is formed of aluminum or an aluminum alloy as in the positive electrode substrate layer of the positive electrode plate, and the negative electrode current collector is formed of copper or a copper alloy as in the negative electrode substrate layer of the negative electrode plate.

The insulating sheet **130** is an insulating sheet-like member that is disposed on the outer surface of the case **110** and covers the outer surface of the case **110**. As the material of the insulating sheet **130**, it is not particularly limited as long as it can ensure the insulation required for the energy storage device **100**, but an insulating resin such as PC, PP, PE, PPS, PET, PBT or ABS resin, an epoxy resin, Kapton, Teflon (registered trademark), silicone, polyisoprene, polyvinyl chloride and the like can be exemplified.

Specifically, the insulating sheet **130** is one insulating sheet disposed so as to cover the entire surface of the case third surface **111***b* of the case **110**, almost the entire surface of the case second surface **111***a* and the case fourth surface **111***c*, and a part of the case first surface **112***a*. In this way, the insulating sheet **130** is disposed on the outer surface of the case **110** with the gas release valve **113** of the lid body **112**, the liquid injection part **114**, and the electrode terminals **120** exposed. The insulating sheet **130** is an example of a second insulating member that covers the case third surface **111***b*.

[3 Detailed Description of Spacer **200**]

Next, the configuration of the spacer **200** is described in detail. FIG. **4** is a perspective view and a cross-sectional view showing the configuration of the spacer **200** according to the present embodiment. Specifically, FIG. **4**(*a*) is a perspective view showing the spacer **200** in a disassembled state and each component together with the joining member **510**. FIG. **4**(*b*) is a cross-sectional view showing the configuration of the upper portion (a portion on the Z-axis plus direction side) when a second member **210** of the spacer **200** of FIG. **4**(*a*) is cut along an IVbc-IVbc line. FIG. **4**(*c*) is a cross-sectional view showing the configuration of the lower portion (a portion on the Z-axis minus direction side). In FIGS. **4**(*b*) and **4**(*c*), components other than the second member **210** are also indicated by broken lines for convenience of description.

As shown in FIG. **4**, the spacer **200** includes a first member **201** and the second member **210**. The first member

201 is a member disposed at a position where it abuts on the case fourth surface **111***c*, which is the side surface (a side surface on the Y-axis direction side) of the energy storage device **100** facing in the Y-axis direction, and includes a first plate portion **220**, a second plate portion **230**, and a joining member **240** that joins the first plate portion **220** and the second plate portion **230**. The second member **210** is a member that is disposed on a side of the first member **201** in a direction intersecting the Y-axis direction and supports the end portion of the first member **201** in a direction intersecting the Y-axis direction. Specifically, it is configured such that at the central position of the second member **210**, an opening **212**, which is a rectangular through hole penetrating in the Y-axis direction, is formed, and the first member **201** is disposed in the opening **212**, and thus, the second member **210** supports the periphery of the first member **201**.

More specifically, as shown in FIG. **10** described later, the second member **210** includes a main body portion **211**, a first protrusion **213**, a second protrusion **214**, and a third protrusion **215**. The main body portion **211** is a rectangular ring-shaped and plate-shaped portion in which the above-described opening **212** is formed at the central position. The main body portion **211** is provided with a spacer first surface **211***a*, a joint protrusion **211***b*, a first concave portion **211***c*, a sandwiching part **211***d*, a second concave portion **211***e*, and a spacer second surface **211***f*.

The spacer first surface **211***a* is an annular flat surface that is disposed on the Y-axis minus direction side of the main body portion **211** and around the opening **212**, and the joining member **510** on the Y-axis minus direction side is disposed thereon. That is, four rectangular joining members **510** are arranged on the spacer first surface **211***a*, and the spacer **200** and the energy storage device **100** on the Y-axis minus direction side are joined together. The spacer second surface **211***f* is an annular flat surface disposed on the opposite side of the spacer first surface **211***a* (Y-axis plus direction side) and around the opening **212**, and the joining member **510** on the Y-axis plus direction side is disposed thereon. That is, four rectangular joining members **510** are arranged on the spacer second surface **211***f*, and the spacer **200** and the energy storage device **100** on the Y-axis plus direction side are joined together.

The joining member **510** is a member that is disposed between the energy storage device **100** and the spacer **200** and joins the energy storage device **100** and the spacer **200**, and is an adhesive layer such as a double-sided tape in the present embodiment. The joining member is not limited to the double-sided tape, and may be an adhesive layer such as an adhesive, and a welded portion when joined by welding, a welded portion when joined by welding, mechanically joined portion by caulking, fitting, or the like, or another joined portion.

The joint protrusion **211***b* is a protruding portion that is long in the X-axis direction and is disposed so as to protrude from the spacer first surface **211***a* and the spacer second surface **211***f* to both sides in the Y-axis direction, and is disposed so as to face the joints **115** of the cases **110** of the energy storage devices **100** on both sides in the Y-axis direction. Specifically, the joint protrusion **211***b* is disposed at a position where it abuts on the joint **115** of the case **110** of the energy storage device **100** when the energy storage device **100** is disposed adjacent to the spacer **200**. That is, the joint protrusion **211***b* abuts on the joint **115** at the time of manufacturing the energy storage apparatus **10**, or abuts on the joint **115** when the case **110** of the energy storage device **100** swells while the energy storage apparatus **10** is in use.

The first concave portion **211***c* is an annular concave portion formed along the periphery of the opening **212** and recessed in the Y-axis plus direction. That is, the second member **210** has the first concave portion **211***c* at the end portion along the opening **212**, and the first member **201** engages with the first concave portion **211***c* in the Y-axis direction. Specifically, the first concave portion **211***c* engages with the engaging part **221** disposed at the end portion of the first plate portion **220** of the first member **201**, so that the first member **201** is disposed with respect to the second member **210**. A configuration may be adopted in which the first member **201** has a first concave portion, and the second member **210** has an engaging part that engages with the first concave portion. That is, one of the first member **201** and the second member **210** may have the first concave portion at the end portion, and the other may have the engaging part **221** that engages with the first concave portion in the Y-axis direction.

The sandwiching part **211***d* is a protrusion that protrudes from the inner wall of the first concave portion **211***c* toward the inside of the opening **212**, and has a function of sandwiching the first member **201**. In the present embodiment, a total of eight sandwiching parts **211***d* are provided on both sides in the X-axis direction and both sides in the Z-axis direction of the first member **201**. That is, two pairs of sandwiching parts **211***d* sandwich the first member **201** in the X-axis direction, and two pairs of sandwiching parts **211***d* sandwich the first member **201** also in the Z-axis direction. Specifically, the sandwiching part **211***d* sandwiches the first member **201** in the X-axis direction and the Z-axis direction by engaging with the end surfaces on both sides in the X-axis direction and both sides in the Z-axis direction of the first member **201**. The shape, number, and arrangement positions of the sandwiching parts **211***d* are not limited to the above.

The second concave portions **211***e* are concave portions recessed in the Y-axis direction from the spacer first surface **211***a* and the spacer second surface **211***f*, are arranged lateral to the first member **201**, and are formed in a long shape along the periphery of the first member **201**. Specifically, the second concave portions **211***e* are arranged at positions adjacent to the sandwiching parts **211***d* in the spacer first surface **211***a*, and are arranged inside (on the first member **201** side) of the second concave portion **211***e* formed in the spacer first surface **211***a* in the spacer second surface **211***f*. The shape, number and arrangement positions of the second concave portions **211***e* are not limited to the above. Instead of the second concave portion **211***e*, a through hole penetrating the main body portion **211** in the Y-axis direction may be formed.

With such a configuration, the sandwiching parts **211***d* and the second concave portions **211***e* are arranged within the range from the spacer first surface **211***a* to the spacer second surface **211***f* in the Y-axis direction. That is, the sandwiching parts **211***d* and the second concave portions **211***e* are formed without a shape protruding from the spacer first surface **211***a* in the Y-axis minus direction, and without a shape protruding from the spacer second surface **211***f* in the Y-axis plus direction. In other words, the spacer first surface **211***a* and the spacer second surface **211***f* are recessed or formed in the same plane as adjacent surfaces at positions overlapping the sandwiching parts **211***d* when viewed in the Y-axis direction. Similarly, the spacer first surface **211***a* and the spacer second surface **211***f* are recessed or formed in the same plane as adjacent surfaces at positions overlapping the second concave portions **211***e* when viewed in the Y-axis direction.

The first protrusion **213** is a long-shaped protruding portion that protrudes toward both sides in the Y-axis direction from the end portion of the main body portion **211** on the Z-axis plus direction side. In the present embodiment, two pairs of first protrusions **213** are arranged side by side in the X-axis direction. The first protrusion **213** is disposed so as to face a convex portion (gasket **121** in the present embodiment) of the energy storage device **100** on the X-axis direction side and protrude along the convex portion. The detailed description of this will be given later.

The second protrusions **214** are protruding portions that protrude from portions on both sides in the X-axis direction and on the Z-axis plus direction side of the main body portion **211** to both sides in the X-axis direction. The second protrusion **214** has a concave-convex portion having at least one of a concave portion and a convex portion formed at the end portion in the X-axis direction. Specifically, the concave-convex portion has a convex portion **214***a* protruding in the Y-axis plus direction, and a concave portion **214***b* formed on the back side of the convex portion **214***a* and recessed in the surface of the end portion of the second protrusion **214** (see FIG. **8**). The detailed description of this will be given later.

The third protrusion **215** is a protruding portion that protrudes from the end portion of the main body portion **211** on the Z-axis minus direction side to the Z-axis minus direction side, which is the direction toward the bottom surface (case third surface **111***b*) side of the energy storage device **100**. The third protrusions **215** are arranged at both end portions of the main body portion **211** in the X-axis direction, and are placed on the insulator **600**. The detailed description of this will be given later.

The second member **210** is formed of an insulating material such as PC, PP, PE, PPS, PET, PEEK, PFA, PTFE, PBT, PES, ABS resin, and a composite material thereof. The second member **210** may be formed of any material as long as it has an insulating property, and all the second members **210** of the plurality of spacers **200** may be formed of members made of the same material, or any of the second members **210** may be formed of members made of different materials.

The first member **201** is formed of a member having higher heat resistance than the second member **210**. Preferably, the first member **201** is formed of a member having higher hardness than the second member **210**. More preferably, the first member **201** is formed of a member having higher heat insulating property than the second member **210**. As the first plate portion **220** and the second plate portion **230** constituting the first member **201** and having high heat resistance and the like, mica plates formed by a dammar material constituted by accumulating and bonding mica pieces (thermal decomposition temperature is, for example, about 600° C. to 800° C.) and the like can be cited. The first plate portion **220** and the second plate portion **230** may be formed of any material as long as it is a member having high heat resistance and the like, and the first plate portion **220** and the second plate portion **230** may be formed of different materials. Since the material and the like of the joining member **240** are the same as the material and the like of the joining member **510**, detailed description thereof will be omitted.

High heat resistance means that it is not easily affected even when exposed to high temperatures, and can maintain physical properties (or maintain shape), and means that it has, for example, a high glass transition temperature, a high deflection temperature under load (heat distortion temperature), or a high melting point. When the glass transition temperature, the deflection temperature under load (heat distortion temperature), and the melting point are compared for two members, and if there are numerical values that are reversed, the member with higher melting point is defined as the member with higher heat resistance. High hardness means that it is hard and difficult to be deformed, for example, it has high Vickers hardness. High heat insulating property means that it is difficult to transfer heat, and means that it has, for example, a low thermal conductivity. The comparison of these heat resistance and the like (heat resistance, hardness and heat insulating property) can be made by appropriately measuring by a known method.

In the above-described embodiment, the first plate portion **220** and the second plate portion **230** are all formed of a member having high heat resistance and the like, but a part thereof may be formed of a member having high heat resistance and the like. The first plate portion **220** and the second plate portion **230** may form a member having high heat resistance and the like by a configuration in which a member (paint or the like) having high heat resistance and the like is disposed (applied) on the surface of a resin substrate.

Specifically, the first plate portion **220** and the second plate portion **230** are, when the energy storage device **100** is disposed adjacent to the spacer **200**, rectangular and flat-plate members arranged at positions abutting on the case fourth surface **111***c* of the energy storage device **100**. That is, the first plate portion **220** and the second plate portion **230** abut on the case fourth surface **111***c* at the time of manufacturing the energy storage apparatus **10**, or abut on the case fourth surface **111***c* when the case **110** of the energy storage device **100** swells while the energy storage apparatus **10** is in use. The first plate portion **220** is formed to have larger widths in the X-axis direction and the Z-axis direction and a larger thickness in the Y-axis direction than the second plate portion **230**.

More specifically, the first plate portion **220** is disposed so as to protrude from the spacer first surface **211***a* toward the energy storage device **100** on the Y-axis minus direction side, and face the central portion of the energy storage device **100** on the Y-axis minus direction side. The second plate portion **230** is disposed on the opposite side (Y-axis plus direction side) of the first plate portion **220**. That is, the second plate portion **230** is disposed so as to protrude from the spacer second surface **211***f* toward the energy storage device **100** on the Y-axis plus direction side and face the central portion of the energy storage device **100** on the Y-axis plus direction side. The first plate portion **220** is disposed so as to protrude to the Y-axis minus direction side relative to the joint protrusion **211***b* on the Y-axis minus direction side, and the second plate portion **230** is disposed so as to be recessed in the Y-axis minus direction relative to the joint protrusion **211***b* on the Y-axis plus direction side. As described above, the first member **201** is formed to have substantially the same thickness as a thickness of the main body portion **211** of the second member **210**, or have a larger thickness than the thickness of the main body portion **211**.

The first plate portion **220** is an example of a central protruding portion, and the central protruding portion (first plate portion **220**) and the joint protrusion **211***b* on the Y-axis minus direction side are an example of a first protruding portion. The second plate portion **230** is also an example of a central protruding portion, and the central protruding portion (second plate portion **230**) and the joint protrusion

211*b* on the Y-axis plus direction side are an example of a second protruding portion. Therefore, the central protruding portion preferably has higher heat resistance and higher hardness than the second member **210**, and more preferably has higher heat insulating property. In other words, each of the first protruding portion and the second protruding portion has a portion having higher heat resistance than the portion having the spacer first surface **211***a* and the spacer second surface **211***f* of the spacer **200**, also preferably has a portion having high hardness, and more preferably has a portion having high heat insulating property.

[4 Detailed Description of Spacer **300**]

Next, the configuration of the spacer **300** is described in detail. FIG. **5** is a perspective view and a cross-sectional view showing the configuration of the spacer **300** according to the present embodiment. Specifically, FIG. **5**(*a*) is a perspective view showing the spacer **300** on the Y-axis plus direction side in FIG. **2** together with the joining members **520** and **530**. FIG. **5**(*b*) is a cross-sectional view showing the configuration of the upper portion (a portion on the Z-axis plus direction side) when the spacer **300** of FIG. **5**(*a*) is cut along a line Vbc-Vbc, and FIG. **5**(*c*) is a cross-sectional view showing the configuration of the lower portion (a portion on the Z-axis minus direction side). In FIGS. **5**(*b*) and **5**(*c*), the joining members **520** and **530** are also indicated by broken lines for convenience of description. The spacer **300** on the Y-axis minus direction side in FIG. **2** also has the same configuration.

As shown in FIG. **5**, the spacer **300** includes a main body portion **310**, a first protrusion **320**, a second protrusion **330**, and a third protrusion **340**. The main body portion **310** is a rectangular and plate-shaped portion, and is provided with a spacer first surface **311**, a first protruding portion **312** having a central protruding portion **312***a* and a joint protrusion **312***b*, and a spacer second surface **313**. The spacer **300** is formed of the same material as the second member **210** of the spacer **200**.

The spacer first surface **311** is an annular flat surface that is disposed on the Y-axis minus direction side of the main body portion **310** and around the central protruding portion **312***a*, and the joining member **520** is disposed thereon. That is, four rectangular joining members **520** are arranged on the spacer first surface **311** to join the spacer **300** and the energy storage device **100** on the Y-axis minus direction side. The spacer second surface **313** is a rectangular flat surface disposed on the opposite side (Y-axis plus direction side) of the central protruding portion **312***a*, and the joining member **530** is disposed thereon. That is, the rectangular joining member **530** is disposed on the spacer second surface **313** to join the spacer **200** and the energy storage device **100** on the Y-axis plus direction side.

The joining member **520** is a member that is disposed between the energy storage device **100** and the spacer **300** to join the energy storage device **100** and the spacer **300**, and the joining member **530** is a member that is disposed between the end member **400** and the spacer **300** to join the end member **400** and the spacer **300**. Since the materials and the like of the joining members **520** and **530** are the same as the materials and the like of the joining member **510**, detailed description thereof will be omitted.

The central protruding portion **312***a* of the first protruding portion **312** is a portion disposed so as to protrude from the spacer first surface **311** toward the energy storage device **100** on the Y-axis minus direction side, and face the central portion of the energy storage device **100** on the Y-axis minus direction side. The joint protrusion **312***b* is a protruding portion that is long in the X-axis direction and is disposed so as to protrude from the spacer first surface **311** to the Y-axis minus direction side, and is disposed so as to face the joint **115** of the case **110** of the energy storage device **100** on the Y-axis minus direction side. Specifically, the central protruding portion **312***a* and the joint protrusion **312***b*, when the energy storage device **100** is disposed adjacent to the spacer **300**, are arranged at positions abutting on the case fourth surface **111***c* and the joint **115** of the case **110** of the energy storage device **100**. That is, the central protruding portion **312***a* and the joint protrusion **312***b* abut on the case fourth surface **111***c* and the joint **115** at the time of manufacturing the energy storage apparatus **10**, or abut on the case fourth surface **111***c* and the joint **115** when the case **110** of the energy storage device **100** swells while the energy storage apparatus **10** is in use. The joint protrusion **312***b* protrudes in the Y-axis minus direction to the same position as the central protruding portion **312***a*.

The first protrusion **320** is a long-shaped protruding portion that protrudes from the end portion of the main body portion **310** on the Z-axis plus direction side to the Y-axis minus direction side. Since the first protrusion **320** has the same configuration as the first protrusion **213** of the spacer **200**, detailed description thereof will be omitted.

The second protrusions **330** are protruding portions that protrude from portions on both sides in the X-axis direction and on the Z-axis plus direction side of the main body portion **310** to both sides in the X-axis direction. The second protrusion **330** has a concave-convex portion having at least one of a concave portion and a convex portion formed at the end portion in the X-axis direction. Specifically, the concave-convex portion has concave portions **331** and **332** recessed in the surfaces at the end portions of the second protrusion **330** (see FIG. **8**). The detailed description of this will be given later.

The third protrusion **340** is a protruding portion that protrudes from the end portion of the main body portion **310** on the Z-axis minus direction side to the Z-axis minus direction side, which is the direction to the bottom surface (case third surface **111***b*) side of the energy storage device **100**. The third protrusion **340** has the same configuration as the third protrusion **215** of the spacer **200**, and thus detailed description thereof will be omitted.

[5 Detailed Description of Insulator **600**]

Next, the configuration of the insulator **600** will be described in detail. FIG. **6** is a perspective view and a plan view showing the configuration of the insulator **600** according to the present embodiment. Specifically, FIG. **6**(*a*) is a perspective view showing the insulator **600** on the X-axis plus direction side in FIG. **2**, and FIG. **6**(*b*) is a perspective view showing the configuration of the insulator **600** of FIG. **6**(*a*) when viewed from the back side. FIG. **6**(*c*) is an enlarged perspective view showing a portion encircled by a broken line in FIG. **6**(*a*) in an enlarged manner. FIG. **6**(*d*) is a plan view showing the configuration of the insulator **600** of FIG. **6**(*b*) when the end portion of the insulator **600** on the Y-axis minus direction side and on the Z-axis plus direction side is seen from the X-axis minus direction side in an enlarged manner. The insulator **600** on the X-axis minus direction side in FIG. **2** also has the same configuration.

As shown in FIG. **6**, the insulator **600** includes an insulator main body portion **610**, an insulator first wall portion **620**, and an insulator second wall portion **630**. The insulator main body portion **610** is a rectangular and plate-shaped portion that is disposed on the X-axis plus direction side of the energy storage device **100** and that extends in the Y-axis direction and is parallel to the YZ plane. The insulator first wall portion **620** is a long and plate-shaped portion that protrudes from the end portion of the insulator main body portion **610** on the Z-axis plus direction side to the X-axis minus direction side and is extended in the Y-axis direction, and is disposed on the Z-axis plus direction side of the energy storage device **100**. The insulator second wall portion **630** is a long and plate-shaped portion that protrudes from the end portion of the insulator main body portion **610** on the Z-axis minus direction side to the X-axis minus direction side and is extended in the Y-axis direction, and is disposed on the Z-axis minus direction side of the energy storage device **100**.

Specifically, the insulator main body portion **610** has a facing portion **611** and an extending portion **612**. The facing portion **611** is a rectangular and plate-shaped portion that is disposed on the X-axis direction side of the case second surface **111***a* so as to face the case second surface **111***a* of the energy storage device **100** and that extends in the Y-axis direction and is parallel to the YZ plane. The extending portion **612** is a portion disposed so as to extend in the Y-axis direction from a portion of the facing portion **611** on the electrode terminal **120** side (Z-axis plus direction side) relative to a portion on the opposite side (Z-axis minus direction side) of the electrode terminal **120** of the facing portion **611**. In other words, the insulator main body portion **610** has a shape in which portions on the Z-axis plus direction side of the end portions on both sides in the Y-axis direction protrude to both sides in the Y-axis direction.

The extending portion **612** has a concave portion **613** and a rib **614**. The concave portion **613** is a notch-shaped concave portion in which the outer edge of the extending portion **612** on the Z-axis minus direction side is recessed in the Z-axis minus direction. The rib **614** is a protruding portion that protrudes from the surface of the extending portion **612**, and is disposed so as to surround the concave portion **613**. Specifically, the rib **614** has a first rib **614***a* extending in the Z-axis direction and a second rib **614***b* extending in the Y-axis direction along the periphery of the concave portion **613**. In the present embodiment, the rib **614** protrudes outward from the outer surface of the extending portion **612**, but it may protrude inward from the inner surface of the extending portion **612**.

A third rib **615** is provided on the inner surface of the insulator main body portion **610**. The third rib **615** is a protruding portion that protrudes inward from the inner surface of the end portion of the facing portion **611** on the Z-axis plus direction side, and is disposed so as to extend in the Z-axis direction. In the present embodiment, a plurality of (eleven) third ribs **615** are arranged side by side at equal intervals in the Y-axis direction.

The insulator first wall portion **620** has first pressing parts **621** and second pressing parts **622** which are a plurality of pressing parts. The plurality of pressing parts (first pressing parts **621** and second pressing parts **622**) are arranged corresponding to each of the plurality of energy storage devices **100**, are portions that press each of the plurality of energy storage devices **100**, and specifically are convex portions that protrude toward the corresponding energy storage devices **100**. Specifically, the insulator first wall portion **620** has at least two first pressing parts **621** and a second pressing part **622** as the plurality of pressing parts. In the present embodiment, two first pressing parts **621** are arranged corresponding to the energy storage devices **100** at both end portions of the plurality of energy storage devices **100**, and a plurality of (ten) second pressing parts **622** are arranged between the two first pressing parts **621**.

Specifically, the first pressing parts **621** and the second pressing parts **622** are convex portions in which the surface of the insulator first wall portion **620** on the Z-axis plus direction side is recessed and the surface of the insulator first wall portion **620** on the Z-axis minus direction side is swollen, and are arranged at equal intervals in the Y-axis direction (alternately arranged with the third ribs **615**). The first pressing part **621** is formed so that a protruding height of the convex portion is higher than that of the second pressing part **622**. That is, a protruding height H**1** of the first pressing part **621** is formed to be larger than a protruding height H**2** of the second pressing part **622** (see FIG. **6**(*d*)). As a result, the first pressing part **621** presses the corresponding energy storage device **100** with a larger force than the second pressing part **622**. The first pressing part **621** may abut on the corresponding energy storage device **100**, and the second pressing part **622** may not abut on the corresponding energy storage device **100**.

The insulator second wall portion **630** is a portion on which the energy storage device **100** and the spacers **200** and **300** are placed. The detailed description of this will be given later.

[6 Detailed Description of Side Plate **700**]

Next, the configuration of the side plate **700** will be described in detail. FIG. **7** is a perspective view showing the configuration of the side plate **700** according to the present embodiment. Specifically, FIG. **7**(*a*) is a perspective view showing the side plate **700** on the X-axis plus direction side in FIG. **2**, and FIG. **7**(*b*) is a perspective view showing the configuration of the side plate **700** of FIG. **7**(*a*) when viewed from the back side. The side plate **700** on the X-axis minus direction side in FIG. **2** also has the same configuration.

As shown in FIG. **7**, the side plate **700** includes a side plate main body portion **710**, a side plate first wall portion **720**, a side plate second wall portion **730**, and a side plate third wall portion **740**.

The side plate main body portion **710** is a rectangular and plate-shaped portion that is disposed on the X-axis plus direction side of the insulator main body portion **610** and that extends in the Y-axis direction and is parallel to the YZ plane. The side plate first wall portions **720** are long and plate-shaped portions that protrude from the end portions on both sides of the side plate main body portion **710** in the Y-axis direction to the X-axis minus direction side and are extended in the Z-axis direction, and are fixed to the end members **400**. The side plate second wall portion **730** is a long and plate-shaped portion that protrudes from the end portion of the side plate main body portion **710** on the Z-axis plus direction side to the X-axis minus direction side and extends in the Y-axis direction, and is inserted and disposed in the insulator first wall portion **620** (see FIG. **10**(*b*)). The side plate third wall portion **740** is a long and plate-shaped portion that protrudes from the end portion of the side plate main body portion **710** on the Z-axis minus direction side to the X-axis minus direction side and extends in the Y-axis direction, and is inserted and disposed in the insulator second wall portion **630** (see FIG. **10**(*c*)).

The side plate main body portion **710** has a concave portion **711** recessed in the outer edge on the Z-axis plus direction side and a concave portion **712** recessed in the outer edge on the Z-axis minus direction side at both end portions in the Y-axis direction. That is, the concave portion **711** is a notch-shaped concave portion in which the outer edge of the side plate main body portion **710** on the Z-axis plus direction side is recessed in the Z-axis minus direction, and has a curved outer edge shape. The concave portion **712** is a notch-shaped concave portion in which the outer edge of the side plate main body portion **710** on the Z-axis minus direction side is recessed in the Z-axis plus direction, and has a curved outer edge shape.

[7 Description of Positional Relationship of Each Component]

Next, the positional relationship among the energy storage device **100**, the spacers **200** and **300**, the end member **400**, the insulator **600**, and the side plate **700** will be described in detail. FIG. **8** is a plan view and a cross-sectional view showing the positional relationship among the energy storage device **100**, the spacers **200** and **300**, the end member **400**, the insulator **600**, and the side plate **700** according to the present embodiment. Specifically, FIG. **8**(*a*) is a plan view of the Y-axis minus direction side of the above-described component when viewed from the Z-axis plus direction, and FIG. **8**(*b*) is a cross-sectional view of a portion of each component of FIG. **8**(*a*) on the X-axis minus direction side and on the Z-axis plus direction side. A portion of each component on the X-axis plus direction side and a portion of each component on the X-axis minus direction side have the same configuration, and a portion of each component on the Y-axis plus direction side and a portion of each component on the Y-axis minus direction side have the same configuration. The same applies to the following.

FIG. **9** is a perspective view showing a positional relationship among the energy storage device **100**, the spacers **200** and **300**, the end member **400**, the insulator **600**, and the side plate **700** according to the present embodiment. Specifically, FIG. **9**(*a*) is a perspective view showing a portion of the above component on the X-axis plus direction side and on the Y-axis minus direction side, and FIG. **9**(*b*) is a perspective view showing the configuration when the side plate **700** is removed from FIG. **9**(*a*).

FIG. **10** is a cross-sectional view showing a positional relationship among the energy storage device **100**, the spacer **200**, the insulator **600**, and the side plate **700** according to the present embodiment. Specifically, FIG. **10**(*a*) is a view of a cross section when the above-described component is cut along the XZ plane, as viewed from the Y-axis minus direction. FIG. **10**(*b*) is an enlarged view of a portion on the X-axis plus direction side and on the Z-axis plus direction side of FIG. **10**(*a*), and FIG. **10**(*c*) is an enlarged view of a portion on the X-axis plus direction side and on the Z-axis minus direction side of FIG. **10**(*a*). FIGS. **10**(*b*) and **10**(*c*) show the configuration when the spacer **200** is removed from FIG. **10**(*a*).

First, as shown in FIG. **8**(*a*), the first protrusions **213** and **320** of the spacers **200** and **300** are arranged so as to protrude along the convex portion (the electrode terminal **120** or the gasket **121**) of the energy storage device **100**. In the present embodiment, the first protrusions **213** and **320** are arranged so as to extend in a long shape along the gasket **121**. That is, the first protrusions **213** and **320** are protrusions that face the X-axis direction side of the gasket **121** and protrude along the gasket **121**. Specifically, the two first protrusions **213** arranged in the X-axis direction are arranged between the two gaskets **121** included in one energy storage device **100** and along the two gaskets **121**. The same applies to the first protrusion **320**. The two first protrusions **213** or the two first protrusions **320** may be arranged at positions sandwiching the two gaskets **121** along the two gaskets **121**.

More specifically, the protrusion amount of the first protrusions **213** and **320** is formed to be smaller than half the width of the case first surface **112***a* of the energy storage device **100** in the Y-axis direction. As a result, the first protrusion **213** and the first protrusion **320**, which are adjacent to each other in the Y-axis direction, or the two first protrusions **213** are arranged so as to face each other and are separated from each other. That is, when two spacers sandwiching the energy storage device **100** are a first spacer and a second spacer, the first spacer has a first protrusion, and the second spacer has another first protrusion, the first protrusion and the other first protrusion are arranged so as to face each other and are separated from each other.

As shown in FIG. **8**(*b*), the spacers **200** and **300** are arranged at positions that do not protrude from the energy storage device **100** in the X-axis direction. That is, the spacers **200** and **300** are arranged inside the energy storage device **100** in the X-axis direction when viewed from the Y-axis direction. In other words, the spacers **200** and **300** are formed so that the width in the X-axis direction is smaller than that of the energy storage device **100**. Alternatively, it can be said that the spacer **200** does not have a portion facing the case second surface **111***a*.

The concave-convex portions formed on the second protrusions **214** and **330** of the spacers **200** and **300** are arranged separated from the energy storage device **100**. Specifically, the concave-convex portion of the second protrusion **214** has a convex portion **214***a* on the Y-axis plus direction side and a concave portion **214***b* on the Y-axis minus direction side of the convex portion **214***a*, and the convex portion **214***a* and the concave portion **214***b* are formed so as to be arranged separated from the energy storage device **100**. More specifically, the convex portion **214***a* is disposed so as to protrude in the Y-axis plus direction along the case corner portion **111***d* of the energy storage device **100**. That is, the convex portion **214***a* is formed so as to protrude in a curved shape toward the case corner portion **111***d* in a state of being separated from the case corner portion **111***d*. The third rib **615** of the insulator **600** is disposed at a position facing the spacer **200** and is inserted into the concave portion **214***b* in the X-axis direction.

Regarding the spacer **300**, the concave-convex portion of the second protrusion **330** has a concave portion **331** on the Y-axis plus direction side and a concave portion **332** on the Y-axis minus direction side, and the concave portions **331** and **332** are formed so as to be arranged separated from the energy storage device **100**.

As shown in FIG. **9**(*a*), the concave portion **711** of the side plate **700** has a shape notched more largely than the concave portion **613** of the insulator **600**, and the insulator **600** is exposed from the concave portion **711**. That is, the concave portion **711** is a concave portion in which the outer edge of the side plate main body portion **710** on the Z-axis plus direction side is recessed relative to the insulator **600**.

Specifically, the concave portion **711** is formed to be largely recessed so that the rib **614** of the insulator **600** is exposed. That is, the rib **614** is disposed in the concave portion **711**. As a result, the first rib **614***a* is disposed on the extending direction side (the Y-axis minus direction side in FIG. **9**) of the extending portion **612** relative to the side plate **700**. The second rib **614***b* is disposed on the electrode terminal **120** side (the Z-axis plus direction side in FIG. **9**) relative to the side plate **700**.

The concave portion **711** is disposed on the X-axis direction side (the X-axis plus direction side in FIG. **9**) of the end member **400**. Therefore, as shown in FIG. **9**(*b*), the extending portion **612**, the concave portion **613**, and the rib **614** of the insulator **600** are also arranged on the X-axis direction side of the end member **400**. Also as shown in FIG. **8**(*b*), the end member **400** includes a first corner portion **410** that is a corner portion on the energy storage device **100** side (the Y-axis plus direction side) and a second corner portion **420** that is a corner portion on the opposite side (the Y-axis minus direction side) of the energy storage device **100**. The first corner portion **410** has a larger radius of curvature at the outer edge than the second corner portion **420**. That is, the first corner portion **410** has a larger rounded outer edge than the second corner portion **420**. The concave portion **711** is formed such that its outer edge has a larger radius of curvature than the first corner portion **410**. That is, the concave portion **711** has a curved outer edge, and the radius of curvature of the curved outer edge is formed to be larger than the radius of curvature of the outer edge of the first corner portion **410**.

As shown in FIGS. **10**(*a*) and **10**(*c*), an end edge of the main body portion **211** of the spacer **200** on the Z-axis minus direction side is disposed on a side opposite to the Z-axis minus direction relative to a surface (the case third surface **111***b*) of the case **110** of the energy storage device **100** on the Z-axis minus direction side. That is, the spacer **200** is disposed so that the main body portion **211** does not protrude from the energy storage device **100** to the Z-axis minus direction side. An end edge of the third protrusion **215** of the spacer **200** on the Z-axis minus direction side is disposed on the same plane (on the same plane P of FIGS. **10**(*a*) and **10**(*c*)) as the case third surface **111***b* of the energy storage device **100**. That is, the spacer **200** is disposed so that the third protrusion **215** also does not protrude from the energy storage device **100** to the Z-axis minus direction side. In other words, the third protrusion **215** has a shape that does not abut on the surface of the energy storage device **100** on the Z-axis minus direction side.

With such a configuration, the surface (the case third surface **111***b*) of the energy storage device **100** on the Z-axis minus direction side is disposed so as to abut on the insulator second wall portion **630** of the insulator **600**. The third protrusion **215** of the spacer **200** is also disposed so as to abut on the insulator second wall portion **630**. That is, since the case third surface **111***b* and the end edge of the third protrusion **215** on the Z-axis minus direction side are arranged on the same plane P, they are placed on the insulator second wall portion **630**. The insulator **600** is disposed so as to abut also on the surface (the case second surface **111***a*) of the energy storage device **100** on the X-axis direction side. The insulator second wall portion **630** is inserted with the side plate third wall portion **740** and is fixed to the energy storage device **100**.

As shown in FIGS. **10**(*a*) and **10**(*b*), the first pressing part **621** and the second pressing part **622** provided on the insulator first wall portion **620** of the insulator **600** press both end portions of the energy storage device **100** in the X-axis direction toward the Z-axis minus direction. The insulator first wall portion **620** is inserted with the side plate second wall portion **730** and is fixed to the energy storage device **100**. As a result, the energy storage device **100** is cooled with the case third surface **111***b* pressed to the cooling device **20**. Since the plurality of energy storage devices **100** are pressed by one insulator **600** and pressed onto the insulator second wall portion **630**, the case third surfaces **111***b* of the plurality of energy storage devices **100** are arranged on the same plane P. As a result, the case third surfaces **111***b* of the plurality of energy storage devices **100** are evenly pressed to the cooling device **20** to be cooled.

Although the spacer **200** has been described with reference to FIG. **10**, the spacer **300** has also a similar configuration.

[8 Description of Effects]

As described above, according to the energy storage apparatus **10** of the present embodiment, the energy storage device **100** has the case first surface **112***a* on which the electrode terminal **120** is disposed, the case second surface **111***a* that is adjacent to the case first surface **112***a* and on the second direction (X-axis direction) side, and the convex portion (gasket **121**) protruding from the case first surface **112***a*, and the spacers **200** and **300** have the first protrusions **213** and **320** that face the second direction side of the convex portion and protrude along the convex portion without having a portion that faces the case second surface **111***a*. In this way, by forming the first protrusions **213** and **320** protruding along the convex portion of the energy storage device **100** on the spacers **200** and **300**, it is possible to position the energy storage device **100** and the spacers **200** and **300** without providing portions that sandwich the side surface of the energy storage device **100** on the spacers **200** and **300**. Thereby, the width of the energy storage apparatus **10** can be prevented from increasing, and the energy storage apparatus **10** can be downsized.

The spacers **200** and **300** are formed so that the widths in the second direction are smaller than that of the energy storage device **100**. By thus forming the widths of the spacers **200** and **300** to be smaller than that of the energy storage device **100**, it is possible to prevent the width of the energy storage apparatus **10** from increasing and to achieve downsizing of the energy storage apparatus **10**.

The first protrusions **213** and **320** of the spacers **200** and **300** are arranged so as to extend in a long shape along the convex portion of the energy storage device **100**. That is, the first protrusions **213** and **320** of the spacers **200** and **300** are not for insulating the case first surface **112***a* of the energy storage device **100**, but for restricting the movement of the convex portion of the energy storage device **100**. Therefore, it is not necessary to increase the width in the second direction, and the first protrusions **213** and **320** can be formed in a long shape. As described above, since it is only necessary to form the long-shaped first protrusions **213** and **320** on the spacers **200** and **300**, it is possible to achieve downsizing of the energy storage apparatus **10**, while easily producing the spacers **200** and **300** and reducing the amount of material used for the spacers **200** and **300**.

The spacers **200** and **300** are formed so that the protrusion amounts of the first protrusions **213** and **320** are smaller than half the width of the case first surface **112***a* of the energy storage device **100** in the first direction (Y-axis direction). By thus reducing the protrusion amounts of the first protrusions **213** and **320** of the spacers **200** and **300**, it is possible to prevent the first protrusions **213** and **320** from interfering with other portions. In addition, by reducing the protrusion amounts of the first protrusions **213** and **320** of the spacers **200** and **300**, it is possible to easily produce the spacers **200** and **300**, reduce the amount of material used for the spacers **200** and **300**, and prevent the first protrusions **213** and **320** from breaking.

Furthermore, the energy storage apparatus **10** includes the spacer **200** (second spacer) that sandwiches the energy storage device **100** with the spacers **200** and **300** (first spacer), and the spacer **200** (second spacer) has another first protrusion **213** that faces the second direction side of the convex portion of the energy storage device **100** and protrudes along the convex portion without having a portion that faces the case second surface **111***a* of the energy storage device **100**. By thus forming the other first protrusion **213** that protrudes along the convex portion of the energy storage device **100** also on the spacer **200** (second spacer), it is possible to position the energy storage device **100** and the spacer **200** (second spacer) without providing a portion that sandwiches the side surface of the energy storage device **100** on the spacer **200** (second spacer). Thereby, the width of the energy storage apparatus **10** can be prevented from increasing, and the energy storage apparatus **10** can be downsized.

Since the first protrusions **213** and **320** of the spacers **200** and **300** (first spacer) and the other first protrusion **213** of the spacer **200** (second spacer) are arranged separated from each other, it is possible to prevent the two first protrusions from interfering with each other. Further, downsizing of the energy storage apparatus **10** can be achieved by using a space, in such a manner that a member such as a thermistor can be disposed in the space formed between the two first protrusions.

In the above description, when the first spacer is used as the spacer **200**, the second spacer can be used not only as another spacer **200** but also as the spacer **300**. In this case, the first protrusion of the second spacer described above becomes the first protrusion **320** of the spacer **300**. The same applies to other descriptions having the same meaning.

The energy storage device **100** has the two convex portions, and the spacers **200** and **300** have two first protrusions **213** and **320** arranged along the two convex portions between the two convex portions or at positions sandwiching the two convex portions. In this way, in the spacers **200** and **300**, by arranging the two first protrusions **213** and **320** at positions sandwiched between the two convex portions of the energy storage device **100**, or at positions sandwiching the two convex portions, it is possible to position the energy storage device **100** and the spacers **200** and **300** in both directions in the arrangement direction of the two first protrusions **213** and **320**.

The convex portion of the energy storage device **100** is the electrode terminal **120** or the gasket **121**, and the first protrusions **213** and **320** of the spacers **200** and **300** are arranged so as to protrude along the electrode terminal **120** or the gasket **121**. That is, the first protrusions **213** and **320** of the spacers **200** and **300** are arranged along the electrode terminal **120** or the gasket **121** of the energy storage device **100**, thereby positioning the energy storage device **100** and the spacers **200** and **300**. As described above, by using the electrode terminal **120** or the gasket **121** of the energy storage device **100** for positioning the energy storage device **100** and the spacers **200** and **300**, it is not necessary to newly form a convex portion, and therefore the configuration is simplified, and downsizing of the energy storage apparatus **10** can be achieved.

In the configuration in which the energy storage device **100** and the spacers **200** and **300** are bonded by the bonding members **510** and **520**, on the spacers **200** and **300**, the first protruding portions that protrude from the spacer first surfaces **211*a*** and **311** on which the bonding members **510** and **520** are arranged toward the energy storage device **100** are provided. Accordingly, when the energy storage device **100** is about to swell, the first protruding portions can suppress the swelling of the energy storage device **100** even if the joining members **510** and **520** are compressed by the swelling force of the energy storage device **100**. Even if the joining members **510** and **520** are not compressed by the swelling force of the energy storage device **100** due to high hardness or the like of the joining members **510** and **520**, the energy storage device **100** may swell toward portions other than the joining members **510** and **520**. However, even in this case, by arrangement of the first protruding portions, it is possible to prevent the energy storage device **100** from swelling toward portions other than the joining members **510** and **520**.

By disposing the first member **201** having high heat resistance at the position abutting on the side surface of the energy storage device **100** on the spacer **200**, it is possible to prevent the spacer **200** from deforming or melting even when the energy storage device **100** has a high temperature. The spacer **200** often has a complicated shape in order to insulate or hold the energy storage device **100**, but it is generally difficult to process a member having high heat resistance into a complicated shape. Therefore, if the second member **210** is formed capable of insulating or holding the energy storage device **100** by configuring the spacer **200** to have the second member **210** that supports the end portion of the first member **201** having high heat resistance, it is not necessary to process the first member **201** into a complicated shape. In particular, since the second member **210** is formed of resin and can be formed into a complicated shape, the second member **210** can be easily formed into a structure that insulates or holds the energy storage device **100**. Accordingly, the first member **201** having high heat resistance can be easily disposed on the spacer **200**. As described above, since it is possible to prevent the spacer **200** from deforming or melting by the first member **201** even when the energy storage device **100** has a high temperature (for example, 600° C. or the like) due to an abnormality of the energy storage device **100**, etc., it is possible to maintain the functions of the spacer **200**, such as swelling suppression or heat insulating property of the energy storage device **100**, and suppress the occurrence of defects.

The spacers **200** and **300** are provided with the third protrusions **215** and **340** having a shape that protrudes to the Z-axis minus direction side of the energy storage device **100** but does not abut on the surface of the energy storage device **100** on the Z-axis minus direction side. The main body portions **211** and **310** of the spacers **200** and **300** are arranged at positions that do not protrude from the surface of the energy storage device **100** on the Z-axis minus direction side by the third protrusions **215** and **340**. As a result, since the surface of the energy storage device **100** on the Z-axis minus direction side can be made to abut on the cooling device **20** without obstruction by the main body portions **211** and **310** and the third protrusions **215** and **340** of the spacers **200** and **300**, the energy storage device **100** can be easily cooled.

By arranging the spacers **200** and **300** inside the energy storage device **100** in the X-axis direction (that is, formed so as not to protrude from the energy storage device **100** in the X-axis direction), the width of the energy storage apparatus **10** in the X-axis direction can be prevented from increasing. By forming concave-convex portions at the end portions of the spacers **200** and **300** in the X-axis direction that are separated from the energy storage device **100**, the creepage distance between the energy storage device **100** and other members (adjacent energy storage device **100** and the like) can be increased at the end portions of the spacers **200** and **300** in the X-axis direction. This makes it possible to reduce the size of the energy storage apparatus **10** while achieving insulation at the end portion of the energy storage device **100** in the X-axis direction.

In the energy storage device **100**, current flows through the electrode terminal **120**. Therefore, in order to ensure insulation between the energy storage device **100** and the conductive member (side plate **700**), it is important to ensure insulation between the electrode terminal **120** of the energy storage device **100** and the conductive member. Therefore, in the first insulating member (insulator **600**) between the energy storage device **100** and the conductive member, the portion on the electrode terminal **120** side is extended. This makes it possible to increase the creepage distance between the electrode terminal **120** of the energy storage device **100** and the conductive member, so that the insulation between the energy storage device **100** and the conductive member can be improved.

In the conductive member, the concave portion **711** that is recessed in the outer edge on the Z-axis plus direction side (electrode terminal **120** side) is formed. This makes it possible to increase the creepage distance between the electrode terminal **120** of the energy storage device **100** and the conductive member, so that the insulation between the energy storage device **100** and the conductive member can be improved.

In the pressing member (insulator **600** and side plate **700**), in order to press all the energy storage devices **100** with a large force, it is necessary to arrange all the energy storage devices **100** so that they are pressed with a large force by all the pressing parts. Since it may be difficult to configure the pressing member, the pressing forces by the pressing parts are made different. As described above, since it is not necessary to press all the energy storage devices **100** with a large force by all the pressing parts, it is possible to easily configure the pressing member that presses the plurality of energy storage devices **100**.

[9 Description of Variants]

Although the energy storage apparatus **10** according to the present embodiment has been described heretofore, the present invention is not limited to the above-mentioned embodiment. That is, the embodiment disclosed this time is an exemplification in all respects and is not restrictive, and the scope of the present invention is shown by the scope of claims, and all modifications within the meaning and scope equivalent to the claims are included.

In the above-described embodiment, the widths of the spacers **200** and **300** in the X-axis direction are made smaller than that of the energy storage device **100**. However, if the spacers **200** and **300** do not have portions facing the case second surface **111***a* of the energy storage device **100**, the widths in the X-axis direction may be the same as that of the energy storage device **100** or larger than that of the energy storage device **100**.

In the above-described embodiment, the convex portion of the energy storage device **100** with which the first protrusions **213** and **320** of the spacers **200** and **300** engage is the electrode terminal **120** or the gasket **121**. However, when the gas release valve **113** or the liquid injection part **114** protrudes, or there is another protruding portion, the protruding portion may be the convex portion.

In the above-described embodiment, the first protrusions **213** and **320** of the spacers **200** and **300** are extended in a long shape in the Y-axis direction, but they may be extended in a long shape in the X-axis direction.

In the above-described embodiment, the first protrusions **213** and **320** of the spacers **200** and **300** are made smaller than half the width of the case first surface **112***a* of the energy storage device **100** in the Y-axis direction. However, the first protrusions **213** and **320** may have a size that is at least half the width of the case first surface **112***a* in the Y-axis direction. In this case, the first protrusion **213** may not have a symmetrical shape in the Y-axis direction, and may have an asymmetrical shape in such a manner that the protrusion of the first protrusion **213** on the Y-axis plus direction side is formed to be smaller than half the width of the case first surface **112***a* in the Y-axis direction, and the protrusion of the first protrusion **213** on the Y-axis minus direction side is formed to be larger than half the width of the case first surface **112***a* in the Y-axis direction. By thus making the lengths of the protrusions different, it is possible to suppress erroneous arrangement of the spacers **200** during manufacturing.

In the above-described embodiment, all the spacers **200** have the above-mentioned configuration, but any spacer **200** may have a configuration different from the above. The same applies to the spacer **300**, the energy storage device **100**, the insulator **600**, and the side plate **700**.

The configurations which are made by arbitrarily combining the respective components which the above-mentioned embodiment and variants thereof include are also included in the scope of the present invention.

The present invention can be realized not only as such an energy storage apparatus **10** but also as the spacers **200** and **300**.

INDUSTRIAL APPLICABILITY

The present invention can be applied to an energy storage apparatus or the like including an energy storage device such as a lithium ion secondary battery.

DESCRIPTION OF REFERENCE SIGNS

10: energy storage apparatus
100: energy storage device
110: case
111*a*: case second surface
111*b*: case third surface
111*c*: case fourth surface
111*d*: case corner
112*a*: case first surface
115: joint
120: electrode terminal
121: gasket
200, **300**: spacer
201: first member
210: second member
211, **310**: main body portion
211*a*, **311**: spacer first surface
211*b*, **312***b*: joint protrusion
211*c*: first concave portion
211*d*: sandwiching part
211*e*: second concave portion
211*f*, **313**: spacer second surface
212: opening
213, **320**: first protrusion
214, **330**: second protrusion
214*a*: convex portion
214*b*, **331**, **332**: concave portion
215, **340**: third protrusion
221: engaging part
240, **510**, **520**, **530**: joining member
312: first protruding portion
312*a*: central protruding portion
600: insulator

The invention claimed is:

1. An energy storage apparatus comprising:

an energy storage device; and

a first spacer disposed adjacent to the energy storage device in a first direction, wherein the energy storage device includes:

a first surface on which an electrode terminal is disposed;

a second surface adjacent to the first surface in a second direction intersecting the first direction; and a convex portion protruding from the first surface,

the first spacer includes a protrusion that faces the convex portion in the second direction and protrudes along the convex portion without having a portion that faces the second surface,

the first direction is a direction in which the energy storage device and the first spacer are arranged,

the energy storage device includes:

a long side surface; and

a short side surface which is smaller than the long side surface and is the second surface, and

the second direction is a direction in which a pair of second surfaces included in the second surface faces,

wherein a whole of the first spacer and a side wall of the energy storage apparatus forms an empty space in the second direction, and

wherein a maximum width of the first spacer in the second direction is smaller than a maximum width of the energy storage device in the second direction, and

wherein the first spacer is a rectangular shape,

wherein the protrusion extends over two energy storage devices that interpose the first spacer.

2. The energy storage apparatus according to claim **1**, wherein the protrusion extends in a long shape along the convex portion.

3. The energy storage apparatus according to claim **1**, wherein a protrusion amount of the protrusion is smaller than half a width of the first surface in the first direction.

4. The energy storage apparatus according to claim **1**, further comprising a second spacer that sandwiches the energy storage device with the first spacer, wherein

the second spacer includes another protrusion that faces the convex portion in the second direction and protrudes along the convex portion without having a portion that faces the second surface.

5. The energy storage apparatus according to claim **1**, wherein

the energy storage device includes two convex portions included in the convex portion, and

the first spacer includes two protrusions included in the protrusion arranged along the two convex portions between the two convex portions or at positions sandwiching the two convex portions.

6. The energy storage apparatus according to claim **1**, wherein

the convex portion is the electrode terminal disposed on the first surface, or a gasket disposed between the electrode terminal and a case of the energy storage device, and

the protrusion protrudes along the electrode terminal or the gasket.

7. The energy storage apparatus according to claim **1**, wherein the first spacer includes a second protrusion protruding in the second direction and an end surface of the second protrusion in the second direction is recessed.

8. An energy storage apparatus comprising:

an energy storage device; and

a first spacer disposed next to the energy storage device in a first direction, wherein the energy storage device includes;

a first surface on which an electrode terminal is disposed;

a second surface adjacent to the first surface in a second direction intersecting the first direction; and

a convex portion protruding from the first surface,

the first spacer includes a protrusion that faces the convex portion in the second direction and protrudes along the convex portion without having a portion that faces the second surface,

wherein a maximum width of the first spacer in the second direction is smaller than a maximum width of the energy storage device in the second direction, and wherein the protrusion extends over two energy storage devices that interpose the first spacer.

9. The energy storage apparatus according to claim **8**, wherein any width of the first spacer in the second direction is smaller than any width of the energy storage device in the second direction.

10. The energy storage apparatus according to claim **8**, wherein the protrusion extends in a long shape along the convex portion.

11. The energy storage apparatus according to claim **8**, wherein a protrusion amount of the protrusion is smaller than half a width of the first surface in the first direction.

12. The energy storage apparatus according to claim **8**, further comprising a second spacer that interposes the energy storage device with the first spacer, wherein

the second spacer includes another protrusion that faces the convex portion in the second direction and protrudes along the convex portion without having a portion that faces the second surface.

13. The energy storage apparatus according to claim **8**, wherein

the convex portion is the electrode terminal disposed on the first surface, or a gasket disposed between the electrode terminal and a case of the energy storage device, and

the protrusion protrudes along the electrode terminal or the gasket.

14. An energy storage apparatus comprising:

an energy storage device; and

a first spacer disposed adjacent to the energy storage device in a first direction, wherein the energy storage device includes:

a first surface on which an electrode terminal is disposed;

a second surface adjacent to the first surface in a second direction intersecting the first direction; and

a convex portion protruding from the first surface,

the first spacer includes a protrusion that faces the convex portion in the second direction and protrudes along the convex portion without having a portion that faces the second surface,

the protrusion extends in a long shape along the convex portion and is brought into contact with the convex portion,

a width of the protrusion in the second direction is smaller than a width of the protrusion in the first direction,

wherein a maximum width of the first spacer in the second direction is smaller than a maximum width of the energy storage device in the second direction, and wherein the protrusion extends over two energy storage devices that interpose the first spacer.

15. The energy storage apparatus according to claim **14**, further comprising a second spacer that interposes the energy storage device with the first spacer, wherein

the second spacer includes another protrusion that faces the convex portion in the second direction and protrudes along the convex portion without having a portion that faces the second surface.

16. The energy storage apparatus according to claim **14**, wherein

the convex portion is the electrode terminal disposed on the first surface, or a gasket disposed between the electrode terminal and a case of the energy storage device, and

the protrusion protrudes along the electrode terminal or the gasket.

17. The energy storage apparatus according to claim **2**, wherein the protrusion is brought into contact with the convex portion.

18. The energy storage apparatus according to claim **8**, wherein both ends of the first spacer in the second direction is devoid of facing to the second surface of the energy storage device in the first direction.

19. The energy storage apparatus according to claim **14**, wherein a maximum width of the first spacer in the second direction is smaller than a maximum width of the energy storage device in the second direction, and

wherein the first spacer is a rectangular shape.

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